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Design, Synthesis, and Investigation of Novel Nitric Oxide (NO)-Releasing Prodrugs as Drug Candidates for the Treatment of Ischemic Disorders: Insights into NO-Releasing Prodrug Biotransformation and Hemoglobin−NO Biochemistry

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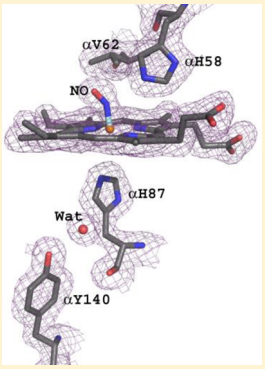
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Supporting Information

ABSTRACT: We have developed novel nitric oxide (NO)-releasing prodrugs of efaproxiral (RSR13) for their potential therapeutic applications in a variety of diseases with underlying ischemia. RSR13 is an allosteric effector of hemoglobin (Hb) that decreases the protein's affinity for oxygen, thereby increasing tissue oxygenation. NO, because of its vasodilatory property, in the form of ester prodrugs has been found to be useful in managing several cardiovascular diseases by increasing blood flow and oxygenation in ischemic tissues. We synthesized three NO-donor ester derivatives of RSR13 (DD-1, DD-2, and DD-3) by attaching the NO-releasing moieties nitrooxyethyl, nitrooxypropyl, and 1-(pyrrolidin-1-yl)diazen-1-ium-1,2-diolate, respectively, to the carboxylate of RSR13. *In vitro* studies demonstrated that the compounds released NO in a time-dependent manner upon being incubated with L-cysteine (1.8−9.3%) or human serum (2.3−52.5%) and also reduced the affinity of Hb for oxygen in whole blood (ΔP_{50} of 4.9−21.7 mmHg vs ΔP_{50} of 25.4−32.1 mmHg for RSR13). Crystallographic studies showed RSR13, the hydrolysis product of the reaction between DD-1 and deoxygenated Hb, bound to the central water cavity of Hb. Also, the hydrolysis product, NO, was observed exclusively bound to the two α hemes, the first such HbNO structure to be reported, capturing the previously proposed physiological bis-ligated nitrosylHb species. Finally, nitrate was observed bound to βHis97. Ultraperformance liquid chromatography−mass spectrometry analysis of the compounds incubated with matrices used for the various studies demonstrated the presence of the predicted reaction products. Our findings, beyond the potential therapeutic application, provide valuable insights into the biotransformation of NO-releasing prodrugs and their mechanism of action and into hemoglobin−NO biochemistry at the molecular level.



Hypoxemia, defined as an insufficient oxygen (O_2) concentration and/or partial pressure of oxygen (pO_2) in arterial blood, is a condition that underlies a myriad of pulmonary, vascular, and blood-related pathologies, including anemia, acute respiratory distress syndrome (ARDS), cardiovascular diseases, congenital heart disease, chronic obstructive pulmonary disease (COPD), emphysema, pneumonia, pneumothorax, pulmonary edema, pulmonary embolism, pulmonary fibrosis, and sleep apnea. Consequently, hypoxemia can often lead to tissue hypoxia or ischemia in essential organs, compromising their physiological function. These conditions can be reversed by increasing or improving the delivery of oxygen to these tissues by various interventions, e.g., by increasing O_2 loads in the lung, enhancing organ perfusion, and/or increasing O_2 unloading in the tissue capillaries, an approach that has proven to be beneficial particularly in attenuating ischemic death resulting from ischemic heart attacks.[1]

Hemoglobin (Hb), the primary molecule responsible for maintaining adequate O_2 levels in the blood and tissues, functions between two allosteric states: the tense state or T-state (unliganded or deoxygenated Hb), which possesses a low O_2 affinity, and the relaxed state or R-state (liganded or oxygenated Hb), which exhibits a high O_2 affinity. The equilibrium between the R- and T-states, which is closely tied to O_2 binding and its release, can be modulated by synthetic

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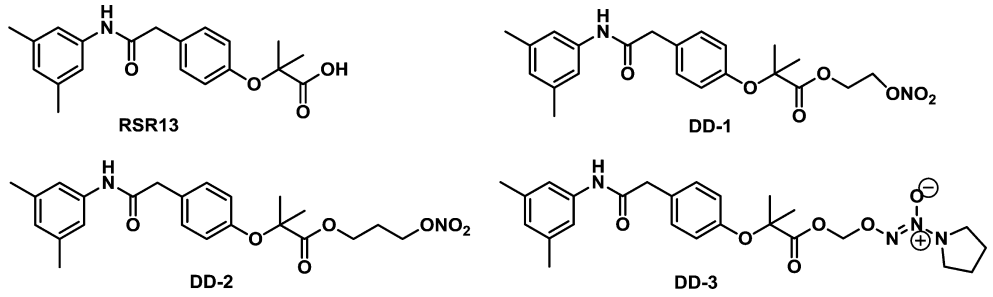


Figure 1. Structures of RSR13 and its NO-releasing derivatives.

allosteric modifiers of Hb.[2] For instance, effectors that shift the oxygen equilibrium curve (OEC) to the left produce R-state/high-O_2 affinity Hb that more readily binds and holds O_2, while shifting the OEC to the right produces T-state/low-O_2 affinity Hb that readily releases O_2. The degree of shift in the OEC is reported as a decrease (left shift) or increase (right shift) in P_{50} (oxygen tension, pO_2, at 50% Hb O_2 saturation). Approximately 25% of the oxygen bound to Hb is released in the tissue capillaries at a physiological pO_2 of ∼20 mmHg during each circulation cycle. The discovery that endogenous 2,3-DPG (2,3-diphosphoglycerate) decreases the O_2 affinity of Hb to allow efficient release of O_2 to tissues prompted the search for synthetic Hb effectors that act like 2,3-DPG but are more potent, i.e., could release more O_2 to tissues. L35 (3,5-trichlorophenylureido-phenoxy-methylpropionic acid), a bezafibrate derivative, was one of the earliest synthetic effectors found to decrease the affinity of Hb for O_2, which in turn stimulates the release of O_2 to tissues.[3] A recent study by Liong and co-workers demonstrated that paramagnetic nanoparticles (PMNPs) coated with L35 (L35-PMNPs) were able to increase targeted tissue pO_2 (using magnetic fields) without inducing any systemic or microhemodynamic changes in hamsters.[4] Efaproxiral or RSR13, 2-(4-{[(3,5-dimethylanilino)-carbon]methyl}phenoxyl)-2-methylpropionic acid (Figure 1), is another synthetic allosteric effector that binds to the central water cavity of Hb and decreases the protein's affinity for O_2, allowing enhancement of O_2 release and delivery of O_2 to ischemic and hypoxic tissues.[2,5] This property has led to this compound being extensively studied or proposed for the treatment of disease states that decrease blood flow to tissues, such as vessel narrowing or weakening of heart pumping capacity, stroke, shock, myocardial ischemia, cerebral traumas, and angina, as well as a radiation enhancer in the radiotherapy of hypoxic tumors.[6−8]

Another molecule that plays a crucial role in regulating the delivery of O_2 to tissues is nitric oxide (NO). In fact, NO is essential in many aspects of human physiology, particularly in cardiovascular homeostasis because of its potent vasodilatory effects.[9−11] It also inhibits vascular remodeling, affecting the multistep cascade of events involved in leukocyte, platelet, and endothelial activation.[12] NO also increases erythrocyte mobility and oxygen delivery in ischemic tissue[13] and acts as a critical mediator of mucosal defense and repair.[14] NO is produced from L-arginine in the vascular endothelium by nitric oxide synthases[15] and from nitrite/nitrate in the erythrocyte by Hb[16] or in the stomach or tongue by sequential reduction of nitrate.[17−20] Because of its critical physiological roles, NO deficiency is implicated in numerous pathological conditions. In particular, reduced local bioavailability of NO in the endothelium is closely associated with coronary artery disease and its risk factors, including diabetes, hypertension, and obesity. Inhaled NO has been shown to reduce the severity and duration of vaso-occlusive crisis in sickle cell patients[21,22] and to provide therapeutic benefits in acute respiratory distress syndrome.[23−25] In addition, long-lived circulating NO-releasing nanoparticles (NO-nps) have been shown to decrease blood pressure, induce vasodilation, and prevent cardiovascular collapse during hemorrhagic events in animal models.[26,27] Compounds with NO releasing abilities, e.g., nitroglycerine, amyl nitrite, isosorbide dinitrate, NO-NSAIDS, nitrosothiols, nicorandil, and sinitrodil, have been studied and/or used for the management of various ischemic syndromes, e.g., cardiovascular disease, pulmonary hypertension, acute lung distress syndrome, etc.[28−36]

Red blood cells (RBCs) and Hb are intimately involved in NO homeostasis, although not without controversy. Studies suggest deoxygenated Hb (deoxyHb) accounts for almost all NO production in RBCs under tissue hypoxia or ischemia or low-blood flow conditions, converting nitrite at the heme moiety to NO and forming methemoglobin (metHb).[16,37−40] Consistently, nitrite has been shown to be vasoactive under physiological conditions.[37,41−44] NO is also capable of reacting directly with the heme of deoxyHb or oxyHb to form nitrosylHb (HbNO), or with the heme of oxyHb to form metHb and nitrate, the latter reaction thought to be a major pathway for removal of excess NO from the body.[45−47]

The crystal structure of deoxyHb in complex with NO coordinated to all four hemes (tetra-ligated T-state HbNO; obtained by exposing deoxyHb crystals to NO gas) has been reported.[48] Richter-Addo and co-workers also recently reported the crystal structure of R-state oxyHb with NO bound to all four heme molecules (tetra-ligated R-state HbNO).[49] In addition to interacting with the hemes of both deoxyHb and oxyHb to form HbNO derivatives, NO also forms a covalent Hb adduct with βCys93 of oxyHb to form S-nitrosoHb (HbSNO)[50−52] that has been proposed as a reservoir for NO. The crystal structure of HbSNO has been reported by the Arnone's group, which was later revised to be actually HbSNOH.[48,53]

In view of the potential beneficial effect of increasing the local NO bioavailability and tissue oxygenation, we have combined the pharmacological effects of RSR13 and NO by derivatizing RSR13 with the NO-releasing groups nitrooxyethyl, nitrooxypropyl, and 1-(pyrrolidin-1-yl)diazen-1-ium-1,2-diolate to form DD-1, DD-2, and DD-3 (Figure 1), respectively. These three compounds were further evaluated *in vitro* for their NO releasing ability, their effects on Hb−O_2 binding, their reaction products in the presence of various biological and nonbiological matrices, and their atomic interaction with Hb.

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EXPERIMENTAL PROCEDURES

Materials and General Procedure. All reagents used in the syntheses, functional assays, and other studies were purchased from Sigma-Aldrich Corp. or Fisher Scientific and used without purification. Routine thin-layer chromatography (TLC) was performed on silica gel GHIF plates (250 μm, 2.5 cm × 10 cm; Analtech Inc., Newark, DE). Column chromatography was performed on silica gel (grade 60 mesh; Bodman Industries, Aston, PA). Melting points were determined on a Fisher Scientific melting point apparatus and were uncorrected. ^{1}H nuclear magnetic resonance (NMR) and ^{13}C NMR spectra were recorded on a Bruker 400 MHz spectrometer, and tetramethylsilane (TMS) was used as an internal standard. Peak positions are given in parts per million (δ). High-resolution mass spectrometry (HRMS) was conducted on a PerkinElmer Flexar UHPLC system with an AxION 2 time of flight (TOF) mass spectrometer, and the molecular weight of the compounds was within 0.05% of calculated values. Infrared spectra were recorded on a Thermo Nicolet iS10 FT-IR instrument. The purity of the compounds was determined by HPLC using a Varian Microsorb 100-5 C18 column (250 mm × 4.6 mm), an acetonitrile (0.1% TFA)/water mobile phase, and a Prostar 325 UV−vis instrument (254 nm) as the detector.

Human blood samples for functional/biological assays were obtained from healthy volunteers at Virginia Commonwealth University (VCU). The use of the biological samples has been reviewed and approved by the IRB, in accordance of institutional regulations. Hemoglobin for functional and/or crystallographic studies was purified from discarded blood or RBCs[54] left over from red cell exchange from the VCU Hospital, Children's Hospital of Philadelphia, or the American Red Cross.

Synthesis of 2-(Nitrooxy)ethyl-2-({4-[2-(3,5-dimethylphenyl)amino]-2-oxoethyl}phenoxy)-2-methylpropanoate, DD-1 (*n* = 1). To stirred concentrated sulfuric acid (98%, 3 mmol) was added concentrated nitric acid (70%, 3 mmol) dropwise at 0 °C under nitrogen. The mixture was stirred for 10 min at the same temperature followed by dropwise addition of 2-bromoethanol (0.31 g, 2.5 mmol) in anhydrous dichloromethane (0.5 mL) while the internal temperature was kept below 5 °C. After 3 h, the reaction was quenched with water (2 mL). Dichloromethane (10 mL) was added, and the organic phase was separated, washed with water (3 mL), dried over Na_2SO_4, and concentrated to give bromoethyl nitrate, which was used without further purification.

RSR13 (0.58 g, 1.7 mmol) and potassium carbonate (0.55 g, 4.0 mmol) were added to a solution of bromoethyl nitrate in anhydrous DMF (4 mL). The reaction mixture was allowed to stir for 24 h at room temperature, diluted with EtOAc (15 mL), washed with water (3 × 4 mL), dried over Na_2SO_4, and evaporated under reduced pressure to give a crude product, which was purified by silica gel column chromatography eluting with hexanes and EtOAc (2:1) to yield 0.7 g of the title compound as a white-colored solid in 95.8% yield: 96.5% HPLC purity (retention time of 4.85 min, 30% H_2O and 70% MeCN over 30 min at 254 nm on a Varian C18 column); mp 98−99 °C; IR (neat) ν_{max} 3294.96, 1741.66, 1637.10, 1507.73, 1272.82, 1141.33 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.20 (d, J = 8.99 Hz, 2H), 7.04 (br s, 3H), 6.85 (d, J = 8.35 Hz, 2H), 6.72 (s, 1H), 4.63 (m, 2H), 4.46 (m, 2H), 3.64 (s, 2H), 2.25 (s, 6H), 1.62 (s, 6H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 174.00, 169.23, 154.86, 138.79, 137.63, 130.54, 128.53, 126.30, 119.81, 117.70, 79.33, 70.19, 61.10, 44.16, 25.52, 21.42; HRMS calcd for $[C_{22}H_{26}N_2O_7Na]^+$ m/z 453.1637, found m/z 453.1710.

Synthesis of 2-(Nitrooxy)propyl-2-({4-[2-(3,5-dimethylphenyl)amino]-2-oxoethyl}phenoxy)-2-methylpropanoate, DD-2 (*n* = 2). To a stirring solution of concentrated sulfuric acid (98%, 3 mmol) was slowly added concentrated nitric acid (70%, 3 mmol) dropwise at 0 °C under nitrogen gas protection. The mixture was stirred for 10 min at the same temperature followed by dropwise addition of 3-bromopropanol (0.35 g, 2.5 mmol) in anhydrous dichloromethane (0.5 mL) while the internal temperature was kept below 5 °C. After the mixture had been stirred for 3 h, the reaction was quenched with water (2 mL). Dichloromethane (10 mL) was added, and the organic phase was separated, washed with water (3 mL), dried over Na_2SO_4, and concentrated to give bromopropyl nitrate, which was used without further purification.

RSR13 (0.58 g, 1.7 mmol) and potassium carbonate (0.55 g, 4.0 mmol) were added to a solution of bromopropyl nitrate in anhydrous DMF (4 mL). The reaction mixture was allowed to stir for 24 h at room temperature, diluted with EtOAc (15 mL), washed with water (3 × 4 mL), dried over Na_2SO_4, and evaporated under reduced pressure to give a crude product, which was purified by silica gel column chromatography eluting with hexanes and EtOAc (2:1) to yield 0.7 g of the title compound as a colorless oil in 92.6% yield: 97.4% HPLC purity (retention time of 5.34 min, 30% H_2O and 70% MeCN over 30 min at 254 nm on a Varian C18 column); IR (neat) ν_{max} 3292.50, 1735.00, 1628.17, 1507.55, 1139.87 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.23 (d, J = 8.60 Hz, 2H), 7.18 (br s, 1H), 7.05 (d J = 8.60 Hz), 6.85 (d, J = 8.35 Hz, 2H), 6.72 (s, 1H), 4.30 (t, J = 6.22 Hz, 2H), 4.25 (t, J = 6.22 Hz, 2H), 3.62 (s, 2H), 2.25 (s, 6H), 2.02 (m, 2H), 1.63 (s, 6H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 174.14, 169.26, 155.00, 138.78, 137.65, 130.46, 128.33, 126.28, 119.11, 117.68, 79.24, 69.52, 61.37, 44.05, 26.28, 25.53, 21.41; HRMS calcd for $[C_{23}H_{28}N_2O_7Na]^+$ m/z 467.1794, found m/z 467.1841.

Synthesis of (*Z*)-2-{[2-({4-[2-(3,5-dimethylphenyl)-amino]-2-oxoethyl}phenoxy)-2-methylpropanoyl]-peroxy}-1-(pyrrolidin-1-yl)diazene 1-Oxide, DD-3. To a DMF (2 mL) solution of freshly prepared O^2-chloromethyl 1-(pyrrolidin-1-yl)diazen-1-ium-1,2-diolate (0.10 g, 0.56 mmol) (prepared as previously reported[34]) was added RSR13 (0.19 g, 0.56 mmol) in one portion, followed by the addition of cesium carbonate (0.18 g, 0.56 mmol). The mixture was stirred at room temperature for 15 h, diluted with water, and extracted with CH_2Cl_2 (3 × 10 mL). The separated organic phase was combined, washed with brine, dried over Na_2SO_4, and concentrated under vacuum to give the crude product, which was further purified by silica gel column chromatography eluting with hexanes and EtOAc (2:1) to yield 0.11 g of the title compound as a colorless oil in 41% yield: 98.7% HPLC purity (retention time of 5.06 min, 30% H_2O and 70% MeCN over 30 min at 254 nm on a Varian C18 column); IR (neat) ν_{max} 3307.34, 2982.95, 1751.57, 1614.43, 1507.17, 1233.15, 1166.75, 1049.51, 939.61 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.17 (d, J = 8.40 Hz, 2H), 7.04 (s, 2H), 6.98 (br s, 1H), 6.86 (d, J = 8.40 Hz, 2H), 6.72 (s, 1H), 5.85 (s, 2H), 3.62 (s, 2H), 3.48 (m, 4H), 2.26 (s, 6H), 1.88 (m, 4H), 1.62 (s, 6H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 172.97, 169.00, 154.70, 138.64, 137.48, 130.29, 128.26, 126.25, 119.93, 117.54, 87.81, 79.12, 50.66, 44.09,

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25.23, 22.91, 21.27; HRMS calcd for $[C_{25}H_{32}N_4O_6Na]^+$ m/z 507.2219, found m/z 507.2241.

Nitric Oxide Release in the Presence of L-Cysteine. Stock solutions of test compounds [10 mM in dimethyl sulfoxide (DMSO)] were prepared, and 60 μL of each compound was added to 5.94 mL of a mixture of 50 mM potassium phosphate buffer (pH 7.4) and methanol [1:1 (v:v)] containing 5 mM L-cysteine. The reaction samples were incubated at 37 °C; 1 mL aliquots were taken at 1, 2, 4, and 8 h and exposed to air for 10 min, and Greiss reagent (250 μL) was added to each solution. The mixture was allowed to stand at 25 °C for 30 min, and the absorbance of each sample was determined at 540 nm. Solutions of 0−100 μM sodium nitrite were used to prepare a nitrite absorbance versus concentration curve under the same experimental conditions. The percent nitric oxide released (quantitated as nitrite ion) was calculated from the standard nitrite concentration curve.

Nitric Oxide Release in the Presence of Human Serum. Stock solutions of test compounds (10 mM in DMSO) were prepared, and 30 μL of each compound was added to 270 μL of human serum. The volume of the mixture, containing 0.2 unit/mL *Aspergillus* nitrate reductase, 50 nM HEPES buffer, 5 μM FAD, and 0.1 mM NADPH, was then increased to 1.5 mL. The reaction mixture was incubated at 37 °C; 250 μL aliquots were taken at 1, 2, 4, and 8 h and 2.5 μL of lactate dehydrogenase (1500 units/mL) and 25 μL of 100 mM pyruvic acid added to oxidize any unreacted NADPH, and the mixture was further incubated for 10 min. Greiss reagent (275.5 μL) was then added to each tube and the mixture allowed to stand at 25 °C for 30 min. Water (555 μL) was added to each reaction mixture; the absorbance was determined at 540 nm, and the percent nitric oxide release was calculated as described above.

Oxygen Equilibrium Curve. The OEC studies with normal whole blood or clarified Hb were conducted using multipoint tonometry or a Hemox Analyzer (TCS Scientific Corp.) as previously reported.[55,56] Stock solutions of RSR13, DD-1, DD-2, and DD-3 were made at a concentration of 250 or 500 mM in DMSO. For the whole blood study, a test compound or DMSO (control) was added to 2 mL of blood (Hemocrit of ∼30%) at a final drug concentration of 2 mM and incubated at 37 °C. At various time points of 1, 2, 4, and 8 h, aliquots were taken and subjected to OEC analysis using multipoint tonometry.

For the OECs of clarified Hb lysates from normal whole blood using a Hemox Analyzer, aliquot samples (500 μL) of Hb lysates were incubated with RSR-13, DD-1, DD-2, DD-3, or DMSO (control) at a final concentration of 2 mM for up to 8 h. At defined intervals, 50 μL aliquots of each sample were mixed with 4.5 mL of 0.1 M potassium phosphate buffer (pH 7.0) at 25 °C in the sample cuvette and subjected to OEC analysis.

Structure of DD-1 in Complex with Deoxygenated Hb. Hb (50 mg/mL) and a 10-fold molar excess of DD-1 were mixed together. A small pellet of sodium dithionite was added and the mixture evacuated for 30 min to make the deoxyHb solution. Subsequent crystallization (under anaerobic conditions) using the batch method as previously described[5,54] was undertaken using a low-salt precipitant of 0.2 M sodium acetate trihydrate, 0.1 M sodium cacodylate trihydrate (pH 6.5), and 30% (w/v) PEG 8000. Dark purple and rectangular crystals grew to a size of approximately 0.2 mm × 0.3 mm × 0.4 mm in 2−4 days. X-ray diffraction data were collected at 100 K with an R-axis IV^{++} image-plate detector using Cu Kα X-rays (λ = 1.54 Å) from a Rigaku Micro-Max-007 X-ray source equipped with varimax confocal optics (Rigaku, The Woodlands, TX) operating at 40 kV and 20 mA. Prior to being used, the crystals were washed in a cryo-protectant solution containing mother liquor (45−55 μL) and glycerol (20 μL) and then flash-frozen. The collected data sets were processed with d*TREK* and the *CCP*4 suite of programs.

The structure was determined by molecular replacement using the T-state native deoxyHb structure [Protein Data Bank (PDB) entry 2DN2] as the search model. One cycle of refinement with CNS and COOT[57,58] identified two symmetry-related bound compounds that appeared to be RSR13 at the central water cavity of the Hb, and two RSR13 molecules were built in the density. The difference map also showed two atom-bound ligands at the α heme distal positions but not the β hemes, and NO molecules were built into the densities. Also included in the model were two nitrate ions bound to the side chains of the two βHis97 residues. Further refinement of the model suggested that the bound effector at the central water cavity could possibly be a mixture of RSR13, DD-1, and RSR13-CH_2CH_2OH. However, in addition to RSR13, only DD-1 was included in the model and only at one of the symmetry-related positions where it was relatively well-defined. Several water molecules and a cacodylate molecule bound at the β-cleft were added, and the structure was refined to final R_{factor} and R_{free} values of 19.8 and 25.3%, respectively. The coordinate has been deposited in the PDB as entry 5E29.

Sample Preparation of DD-1 and DD-3 with Buffer or DMSO for Ultraperformance Liquid Chromatography−Mass Spectrometry (UPLC−MS) Analysis. Twenty microliters of the test compound (20 mM in DMSO) was mixed with 480 μL of buffer [50 mM Tris buffer containing 0.5 mM EDTA (pH 8.6)], the mixture incubated at room temperature or 37 °C for 1 day, and the sample filtered prior to UPLC−MS analysis. Another DMSO sample preparation without the buffer was also used for UPLC−MS analysis after incubation at 37 °C for 1 day and diluted with 500 μL of acetonitrile.

Sample Preparation of DD-1 and DD-3 with L-Cysteine or Serum for UPLC−MS Analysis. Twenty microliters of the test compound (20 mM in DMSO) was mixed with 480 μL of buffer containing L-cysteine [50 mM phosphate buffer (pH 7.4) and MeOH in a 1:1 ratio containing 5 mM L-cysteine] or with 90 μL of serum, 25 μL of 1 M HEPES buffer, and 365 μL of water. The mixture was then incubated at room temperature or 37 °C for 1 day and the sample filtered prior to UPLC−MS analysis.

Sample Preparation of DD-1 with a Solution Mimicking the Crystallization Condition for UPLC−MS Analysis. DD-1 (4 mg) dissolved in 100 μL of DMSO was added to 1 mL of the crystallization buffer [0.2 M sodium acetate trihydrate, 0.1 M sodium cacodylate trihydrate (pH 6.5), and 30% (w/v) PEG 8000], followed by addition of ∼1 mg of sodium dithionite. The mixture was incubated at room temperature for 2 or 8 days and the sample filtered prior to LC−MS analysis.

Sample Preparation of DD-1 and DD-3 with Clarified Hb, Whole Blood, or Bovine Serum Albumin for UPLC−MS Analysis. Ten microliters of both a DD-1 solution and a DD-3 solution (20 mM in DMSO) were each mixed with either 90 μL of hemoglobin [2 mM in 50 mM Tris buffer containing 0.5 mM EDTA (pH 8.6)], 90 μL of whole blood (30% HCT), or 90 μL of BSA (2 mM in 50 mM Tris buffer and 0.5 mM

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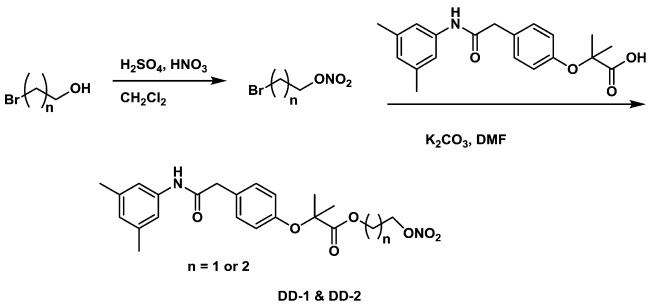
EDTA) and incubated at room temperature or 37 °C for 16 h; 500 μL of acetonitrile was added to each sample to precipitate the protein constituents. After centrifugation, the supernatant solutions were subjected to UPLC−MS analysis.

General UPLC−MS Protocol. UPLC−MS analysis was performed using a Waters Acquity H-Class UPLC system connected to a PDA detector and a Waters Acquity TQD detector. A Waters Acquity UPLC BEH C18 column (1.7 μm, 2.1 mm × 50 mm) was used with a Vanguard precolumn attached. Solvent A consisted of a 90:10 water/acetonitrile mixture with 0.01% formic acid and 0.1 mM hexylamine, while solvent B consisted of a 90:10 acetonitrile/water mixture with 0.01% formic acid and 0.1 mM hexylamine. A gradient run was performed such that the level of solvent B was increased from 0 to 100% from 0 to 7 min, followed by a 5 min wash at 100% B and then a return and re-equilibration at 100% A over the next 3 min. Ten microliters of the sample was injected per run. The eluent of the column was connected to a PDA UV detector that scanned from 250 to 350 nm, and a two-dimensional channel of 280 nm was set. The eluent was then introduced into the TQD detector. The TQD detector was set to positive ionization mode with a capillary voltage of 3.20 kV, a cone voltage of 20 V, an extractor voltage of 3 V, and a RF lens voltage of 0.1 V. The source temperature was set at 150 °C, and the desolvation temperature was set at 350 °C; the desolvation and cone gas flows were set at 650 and 50 L/h, respectively. Scans were made from m/z 100 to 700 with a scan duration of 0.5 s to obtain mass spectra at different time points.

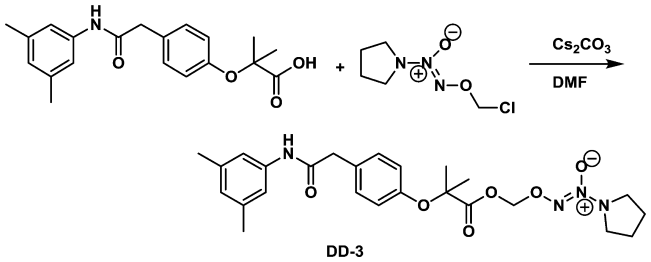
RESULTS AND DISCUSSION

Chemistry. In this study, we incorporated the well-established NO-donor group, nitrooxyalkyl or 1-(pyrrolidin-1-yl)diazen-1-ium-1,2-diolate,[33] onto RSR13, resulting in compounds DD-1, DD-2, and DD-3 (Figure 1). The synthetic routes for the preparation of the nitrooxyalkyl derivatives, DD-1 and DD-2, and the diazen-1-ium-1,2-diolate derivative, DD-3 are shown in Schemes 1 and 2, respectively. For compounds DD-1 and DD-2, the synthesis was achieved by coupling RSR13 with bromoethyl nitrate and bromopropyl nitrate, respectively, which were prepared from the nitration of bromoethanol and bromopropanol, respectively. Compound DD-3 was prepared through the coupling of RSR13 with O^2-chloromethyl 1-(pyrrolidin-1-yl)diazen-1-ium-1,2-diolate, which was synthesized utilizing a published method.[59]

Scheme 1. Synthesis of Compounds DD-1 and DD-2



Scheme 2. Synthesis of Compound DD-3



The Prodrugs Are Capable of Releasing NO. The key requirement for any therapeutic effectiveness of the NO-releasing ester prodrugs is their ability to undergo biotransformation *in vivo*, either enzymatically or nonenzymatically, to release the parent RSR13 and NO. We therefore first evaluated DD-1, DD-2, and DD-3 *in vitro* for their NO releasing ability with both L-cysteine and serum. At various incubation times (1, 2, 4, and 8 h) for the compounds with L-cysteine or human serum at 37 °C, aliquots were added to Griess reagent to quantify the released NO in the form of nitrite ion (NO_2^-) as previously reported.[34,60] Results are expressed as a percentage of NO released relative to the theoretical maximal release of 1 mol of NO/mol of DD-1 or DD-2, and 2 mol of NO/mol of DD-3, and are shown in Table 1 and Figure 2. With L-cysteine, after 1 h, the three tested compounds, DD-1, DD-2, and DD-3, resulted in %NO releases of 3.8, 3.5, and 1.8, respectively, which gradually increased to 9.3, 5.3, and 5.1, respectively, at 8 h (Figure 2A).

With serum (containing esterases),[34] formation of nitrite with DD-1 was almost constant at ∼9.5% during the 8 h experiment, while formation of DD-2 increased from 2.3 to 7.0% at 2 h and decreased to 5.1% at 8 h (Figure 2B). In contrast, the diazen-1-ium-1,2-diolate prodrug, DD-3, showed substantial nitrite formation at 1 h (41.5%) in serum, which increased to 50.6% at 8 h (Figure 2B). As expected, RSR13 did not show any formation of nitrite (not shown in Table 1).

In general, the nitrooxyalkyl prodrugs, DD-1 and DD-2, appear to produce more NO and to do so more rapidly than the diazen-1-ium-1,2-diolate prodrug, DD-3 with L-cysteine, while the opposite is true with serum, consistent with previous findings for these two classes of NO-releasing moieties.[33] Among the nitrooxyalkyl prodrugs, DD-1 (with the shorter nitrooxyalkyl chain) released more NO than DD-2 (with the longer nitrooxyalkyl chain) with both L-cysteine and serum, especially in the former matrix. These findings indicate that *in vivo* (serum) rapid, enzymatic ester hydrolysis is the major mechanism of NO release in blood, favoring DD-3, while the chemical release of NO from L-cysteine favors the nitrooxyalkyl prodrugs, DD-1 in particular. A notable pharmacologic advantage of diazen-1-ium-1,2-diolate prodrugs is that their use is not likely to lead to clinical nitrate tolerance development compared to that of the nitrooxyalkyl prodrugs, because NO release by the latter is dependent on metabolic activation that decreases in efficiency upon continued use.[34,61−63] These observations suggest that *in vivo*, the diazen-1-ium-1,2-diolate prodrug may release more NO than the nitrooxyalkyl prodrugs, i.e., have a NO-releasing pharmacologic property superior to that of their nitrooxyalkyl prodrug counterparts.

The Prodrugs Decrease the Affinity of Hb for O_2. Biotransformation of the DD compounds is anticipated to release RSR13, which *in vivo* is expected to bind to Hb and decrease O_2 affinity. We therefore conducted time-dependent (1, 2, 4, and 8 h) *in vitro* OEC studies to determine the effect of RSR13, DD-1, DD-2, and DD-3 at concentrations of 2 mM on the Hb affinity for O_2 in normal human blood (Hemocrit of ∼30%) at 37 °C using multipoint tonometry as previously

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Table 1. NO Release (percent of theoretical maximum) from DD-1, DD-2, and DD-3 in the Presence of L-Cysteine or Human Serum[a]

		%NO release			
compound	matrix	1 h	2 h	4 h	8 h
DD-1	L-Cys	3.8 ± 0.4	5.9 ± 0.3	8.8 ± 0.4	9.3 ± 0.3
	serum	10.0 ± 0.1	9.7 ± 0.3	8.5 ± 1.0	10.4 ± 0.2
DD-2	L-Cys	3.5 ± 0	4.2 ± 0.1	5.2 ± 0.4	5.3 ± 0.3
	serum	2.3 ± 2.0	7.0 ± 1.8	5.8 ± 0	5.1 ± 0.04
DD-3	L-Cys	1.8 ± 0.07	3.1 ± 0.2	4.7 ± 0.07	5.1 ± 0.1
	serum	41.5 ± 0.2	45.7 ± 1.4	50.6 ± 1.9	52.5 ± 0.5

[a]The experiments were performed in duplicate or triplicate. All compounds were solubilized in DMSO (2%). Control experiments without test compound also included 2% DMSO.

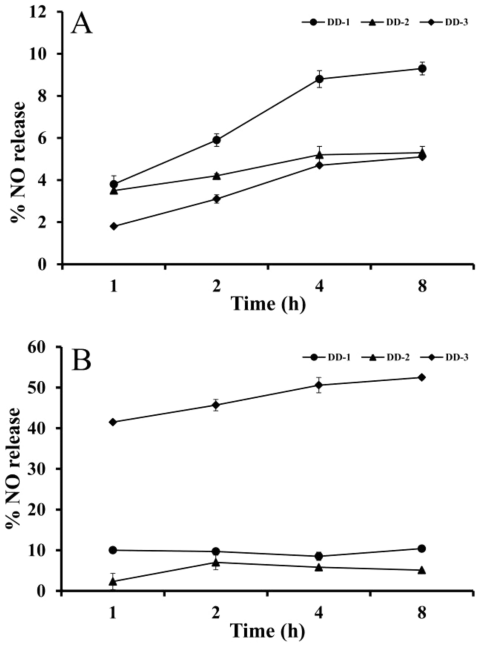


Figure 2. %NO release by (●) DD-1, (▲) DD-2, and (◆) DD-3: (A) L-cysteine and (B) serum.

described.[55,56] Results of the whole blood assay are listed in Table 2 and also shown in Figure 3A. The ΔP_{50} of RSR13

Table 2. Effects of Compounds on Hb Oxygen Affinity with Normal Whole Blood (30% Hematocrit)[a]

	ΔP_{50} (mmHg)			
compound	1 h	2 h	4 h	8 h
RSR13	25.4 ± 4.5	32.1 ± 5.8	30.4 ± 3.1	26.1 ± 2.7
DD-1	10.0 ± 0.5	15.2 ± 0.8	13.6 ± 0.5	11.5 ± 3.1
DD-2	6.3 ± 0.9	9.5 ± 1.7	13.0 ± 0.4	20.7 ± 0.8
DD-3	4.9 ± 0.9	10.4 ± 0.4	20.2 ± 2.7	21.7 ± 5.6

[a]The experiments were performed in duplicate or triplicate. $\Delta P_{50} = P_{50}$ control − P_{50} sample. All compounds were solubilized in DMSO (2%). Control experiments without test compound also included 2% DMSO.

increased from 25.4 mmHg at 1 h to a maximum of 32.1 mmHg at 2 h and then gradually decreased to 26.1 mmHg at 8 h. DD-1 increased the P_{50} by 10 mmHg at 1 h, peaking at 2 h with a ΔP_{50} of 15.2 mmHg, and gradually decreasing to 11.5 mmHg at 8 h. DD-2 and DD-3 showed ΔP_{50} values of 6.3 and 4.9 mmHg, respectively, at 1 h that gradually increased and peaked at 8 h with ΔP_{50} values of 20.7 and 21.7 mmHg,

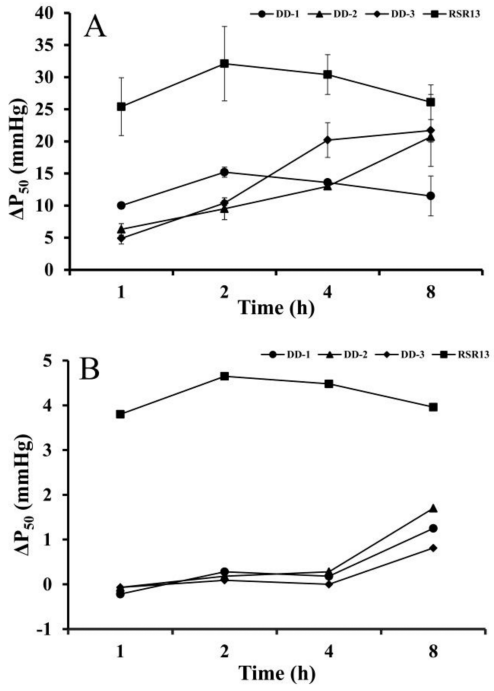


Figure 3. Decrease in Hb O_2 affinity (increase in P_{50}) by (■) RSR13, (●) DD-1, (▲) DD-2, and (◆) DD-3: (A) whole blood (30% Hematocrit) and (B) clarified Hb (2 mM).

respectively. At their peak effects, RSR13 (ΔP_{50} of 32.1 mmHg at 2 h) is the most potent, followed by DD-3 and DD-2 (ΔP_{50} of ~21 mmHg at 8 h), and finally DD-1 (ΔP_{50} of ~15 mmHg at 2 h). It appears from the OEC study that the allosteric property of the diazen-1-ium-1,2-diolate prodrug is relatively superior to those of its nitrooxyalkyl prodrug counterparts.

We further evaluated the allosteric activity of RSR13, DD-1, DD-2, and DD-3 (at a concentration of 2 mM) with clarified Hb (devoid of any other proteins, enzymes, plasma, or cell membranes), also in a time-dependent fashion (Table 3 and Figure 3B). For the first 4 h, there was very little, if significant at all, decrease in Hb O_2 affinity in the presence of the prodrugs; however, at 8 h, DD-1, DD-2, and DD-3 showed decreases in Hb O_2 affinity of 1.25, 1.7, and 0.81 mmHg, respectively. These results contrast with the value for RSR13, which maintained an increase in the P_{50} shift of ~4 mmHg during the 8 h experiment (Table 3 and Figure 3B). Clearly, whole blood, plasma esterases, and possibly other enzymes are able to hydrolyze the carboxylate ester group of the prodrugs releasing RSR13, which is expected to bind to the central water cavity of the Hb, stabilizing the T-state and decreasing the affinity of the protein

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Table 3. Effects of Compounds on Hb Oxygen Affinity with Clarified Hb (2 mM)[a]

	ΔP_{50} (mmHg)			
compound	1 h	2 h	4 h	8 h
RSR13	3.8	4.65	4.48	3.96
DD-1	−0.22	0.28	0.18	1.25
DD-2	−0.07	0.18	0.28	1.70
DD-3	−0.07	0.09	0.0	0.81

[a]$\Delta P_{50} = P_{50}$ control − P_{50} sample. All compounds were solubilized in DMSO (2%). Control experiments without test compound also included 2% DMSO.

for oxygen.[2,5] In contrast, such a hydrolytic process may not occur with clarified Hb to release RSR13, consistent with UPLC−MS studies (*vide infra*). Thus, the observed allosteric effect with free Hb, albeit delayed, is most likely due to direct prodrug binding to Hb. The data also suggest that unlike RSR13, these prodrugs react very slowly and weakly with Hb, consistent with the crystallographic and UPLC−MS studies (*vide infra*).

Structure of Deoxygenated Hb Cocrystallized with DD-1. We have determined the crystal structure of deoxygenated Hb that was cocrystallized with DD-1 to 1.85 Å resolution. To obtain the crystals, deoxygenated Hb (made by adding a small pellet of sodium dithionite to Hb) was incubated with DD-1, and the complex solution was crystallized using a low-salt precipitant [0.2 M sodium acetate trihydrate, 0.1 M sodium cacodylate trihydrate (pH 6.5), and 30% (w/v) PEG-8000] and the batch method, all under anaerobic conditions as previously described.[5,54] The structure was determined using molecular replacement with the high-salt T-state deoxyHb structure (PDB entry 2DN2).[64] Detailed crystallographic data, including data collection and refinement statistics, are summarized in Table 4. The final tetrameric Hb structure contained a pair of RSR13 molecules and one molecule of DD-1 (RSR13-$CH_2CH_2ONO_2$), all bound at the central water cavity (Figure 4). Another DD-1 hydrolysis product, RSR13-CH_2CH_2OH, might also be one of the central water cavity-bound ligands, although it was not included in the refinement. We also observed two NO molecules bound to the α heme irons, two nitrate molecules bound to the βHis97 (β1His97 and β2His97) imidazoles, and a cacodylate molecule (from the precipitant used for crystallization) bound to the dyad axis of the β-cleft (Figure 4). As will be shown later, the sodium dithionite used to deoxygenate the Hb for crystallization most likely led to formation of RSR13 and RSR13-CH_2CH_2OH, and apparently the NO−Hb adducts. The structure reported here will hence be termed bis-ligated T-state HbNO.

The quaternary structure of the bis-ligated T-state HbNO is similar to the previously published T-state Hb structure (PDB entry 2DN2, rmsd of 0.45 Å) or the tetra-ligated T-state HbNO structure (PDB entry 1RPS, rmsd of 0.38 Å); however, as will be discussed later, there are several notable localized structural differences between these structures.

DD-1 and Its Hydrolysis Products Bind at the Central Water Cavity of DeoxyHb. DD-1 appears to bind to the central water cavity of deoxyHb as a mixture of DD-1 (RSR13-$CH_2CH_2ONO_2$), RSR13-CH_2CH_2OH, and RSR13. During the initial refinements of the structure, we modeled a pair of RSR13 molecules in a symmetry-related manner into the difference electron density map at the central water cavity of Hb (Figure 5A) as previously reported.[5] Subsequent refinement of the model suggested the ligand density (Figure 5A,B) to be actually a mixture of DD-1 and RSR13, and possibly RSR13-CH_2CH_2OH, consistent with UPLC−MS analysis (*vide infra*). Two molecules of RSR13, bound in a symmetry-related fashion to the α1α2β1 and α2α1β2 subunits, and one molecule of DD-1 occupying almost the same position as the RSR13 α1α2β1 bound molecule (each refined at 40% occupancy) were modeled and refined. There was no observable density for the nitrooxyethyl moiety at the α2α1β2 binding site, and therefore, only RSR13 was refined at this site at 80% occupancy. Thus, despite the large amount of DD-1 used for the cocrystallization experiment (10-fold molar excess vs Hb), the bound compounds showed relatively weak density. This observation is in contrast to our previous structural study of deoxyHb in complex with RSR13 (1:5 molar ratio) that showed very strong RSR13 electron density.[5] Notably, other analogues of RSR13 that have amino acid substitution at the carboxylate are also characterized by weak density and/or allosteric potency significantly lower than that of RSR13.[65,66] As will be discussed later, UPLC−MS analysis of the crystallization solution showed DD-1 to predominate the mixture, which is in at least 3-fold excess over RSR13 and RSR13-CH_2CH_2OH combined, suggesting that the binding site may have a weaker affinity

Table 4. Crystallographic Data and Refinement Statistics for Bis-ligated T-State HbNO[a]

	DD-1
Data Collection	
space group	$P2_12_12$
unit cell parameters	95.88 Å, 98.42 Å, 65.44 Å, 90.00°, 90.00°, 90.00°
no. of molecules in the asymmetric unit	1
resolution (Å)	29.54−1.85 (1.92−1.85)
no. of measurements	291228 (19980)
no. of unique reflections [$I/\sigma(I)$]	52662 (4994)
average redundancy	5.53 (4.00)
completeness (%)	98.2 (94.3)
average redundancy	5.53 (4.0)
R_{merge}[b] (%)	8.4 (31.3)
average $I/\alpha(I)$	12.3 (4.5)
Refinement	
resolution limits (Å)	29.54−1.85 (1.92−1.85)
no. of reflections	52628 (4991)
R factor (%)	19.8 (37.3)
R_{free}[c] (%)	25.3 (37.8)
rmsd from standard geometry	
bond lengths (Å)	0.10
bond angles (deg)	1.4
dihedral angles	
preferred (%)	95.92
allowed regions (%)	4.08
average B factor (Å)	
all atoms	25.5
protein	22.5
heme/NO	20.5/18.6
water/cacodylate	35.4/75.4
DD-1/RSR13	37.0
NO_3	46.4

[a]Values in parentheses are those of the outermost resolution bin. [b]$R_{merge} = \sum_{hkl}\sum_i/I_{hkli} - \langle I_{hkli}\rangle/\sum_{hkl}\sum_i\langle I_{hkli}\rangle$. [c]$R_{free}$ was calculated with 5% of the reflections excluded from the refinement.

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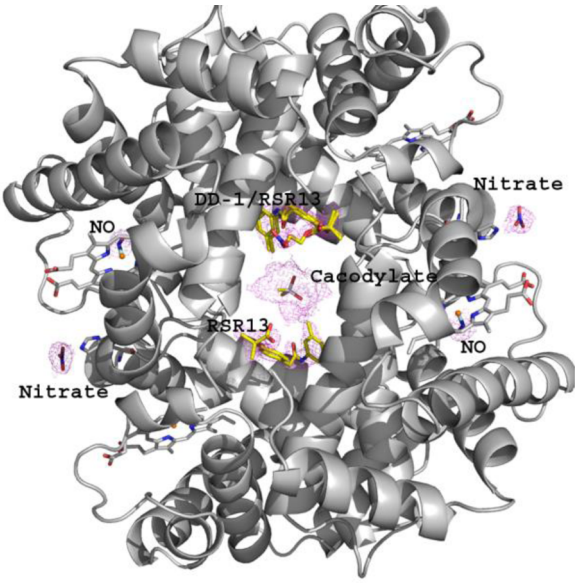


Figure 4. Bis-ligated T-state HbNO structure (ribbon) showing the final $2F_o - F_c$ electron electron density map (contoured at 1.0σ) of bound DD-1 (yellow sticks) and its hydrolysis products, RSR13 (yellow sticks), NO (blue sticks), and nitrate (blue sticks), as well as cacodylate (yellow sticks) from the precipitant.

for DD-1 and thus explaining the weak presence of the effector in the structure. This observation is also the most likely explanation for why the NO-releasing prodrugs showed a very weak and delayed allosteric effect upon being incubated directly with clarified Hb.

The mode of interaction of RSR13 with Hb and the mechanism underlying RSR13 allosteric activity have been well-elucidated.[2,5] Except for additional interactions by the nitro-oxyethyl moiety of DD-1, the almost superposed compounds, DD-1 and RSR13, bind deoxyHb in a similar fashion, interacting with the $\alpha 1$, $\alpha 2$, and $\beta 1$ subunits[5] (Figure 5). Only RSR13 was modeled at the opposite $\alpha 2\alpha 1\beta 2$ binding site. At the $\alpha 1\alpha 2\beta 1$ binding site, the carboxylate of RSR13 or the ester carbonyl oxygen of DD-1 forms an intersubunit water-mediated hydrogen bond interaction with the amine of $\alpha 2$Arg141 (Figure 5C). The methylpropionate and phenyl groups of DD-1 or RSR13 make several intra- and intersubunit hydrophobic interactions with $\beta 1$Tyr35, $\beta 1$Trp37, $\beta 1$Leu105, $\alpha 2$Thr137, $\alpha 2$Tyr140, and $\alpha 2$Pro95, while the amide oxygen or nitrogen makes direct or water-mediated intrasubunit hydrogen bond interactions with the side chains of $\alpha 1$Lys99, $\beta 1$Tyr35, and $\beta 1$Asn108. The dimethylbenzene ring makes hydrophobic intrasubunit interactions with $\alpha 1$Phe36, $\alpha 1$Lys99, $\alpha 1$His103, $\alpha 1$Leu100, and $\beta 1$Asn108. The side chain of $\beta 1$Asn108 is also engaged in hydrogen bond interaction with the π electrons of the dimethylbenzene ring. The nitrate moiety of DD-1 is further involved in direct or water-mediated intrasubunit hydrogen bond interactions with $\alpha 1$Thr134, $\alpha 1$Ala30, and $\alpha 1$Ser131. As noted above, we did not model DD-1 at the $\alpha 2\alpha 1\beta 2$ RSR13 binding site because of weak density where the nitrooxyethyl moiety was supposed to be; rather, water molecules were placed in those densities. Nonetheless, we observe similar RSR13 symmetry-related interactions at the $\alpha 2\alpha 1\beta 2$ binding site as described above for $\alpha 1\alpha 2\beta 1$-bound RSR13. The DD-1−RSR13 interactions, as previously noted for RSR13 and several of its analogues,[2,5,65,66] serve to tie the deoxyHb subunits together, stabilizing the T-state, reducing allosteric interactions among the subunits, and decreasing the affinity of Hb for O_2. The structural study also suggests that *in vivo* biotransformation of DD-1 to yield RSR13 would likely result in binding of RSR13 to the protein to affect its allosteric activity, consistent with the OEC studies. Although we did not determine the structures of deoxyHb in complex with DD-2 or DD-3, we expect the respective structures to show bound RSR13, NO, and nitrate molecules as described above for the DD-1 structure. We also expect intact DD-2 and DD-3 to be able to bind to the allosteric site as described for DD-1; however, the bulky nature of the DD-3 (pyrrolidin-1-yl)diazen-1-ium-1,2-diolate moiety would introduce steric interaction and thus decrease its affinity for the protein compared to that of DD-1 or DD-2. Nonetheless, direct binding of all the prodrugs to the Hb is expected to be slow and weak as observed with the OEC study with clarified Hb.

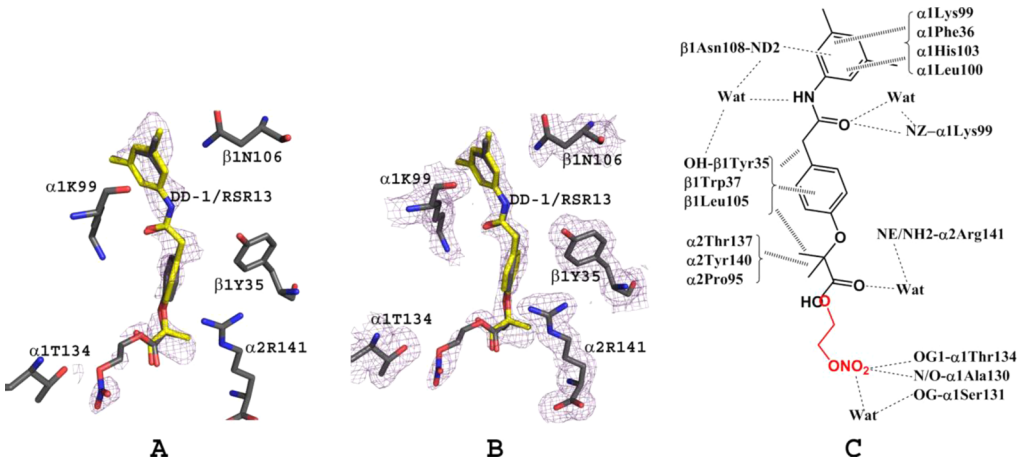


Figure 5. DD-1−RSR13 interactions in the central water cavity ($\alpha 1\alpha 2\beta 1$ binding site) of bis-ligated T-state HbNO. Note that RSR13 (yellow) and DD-1 (gray) occupy the same site of the $\alpha 1\alpha 2\beta 1$ binding site, and each was refined with 40% occupancy. (A) Difference electron density ($F_o - F_c$) map of bound DD-1 and RSR13 (before DD-1 and RSR13 were built into the model), contoured at 2.5σ. (B) Final $2F_o - F_c$ map of bound DD-1 and RSR13, contoured at 1.0σ. The two maps are superimposed with the final refined DD-1−RSR13 model. For the sake of clarity, not all interacting residues are shown. (C) Schematic representation of interactions between the DD-1−RSR13 model and the protein.

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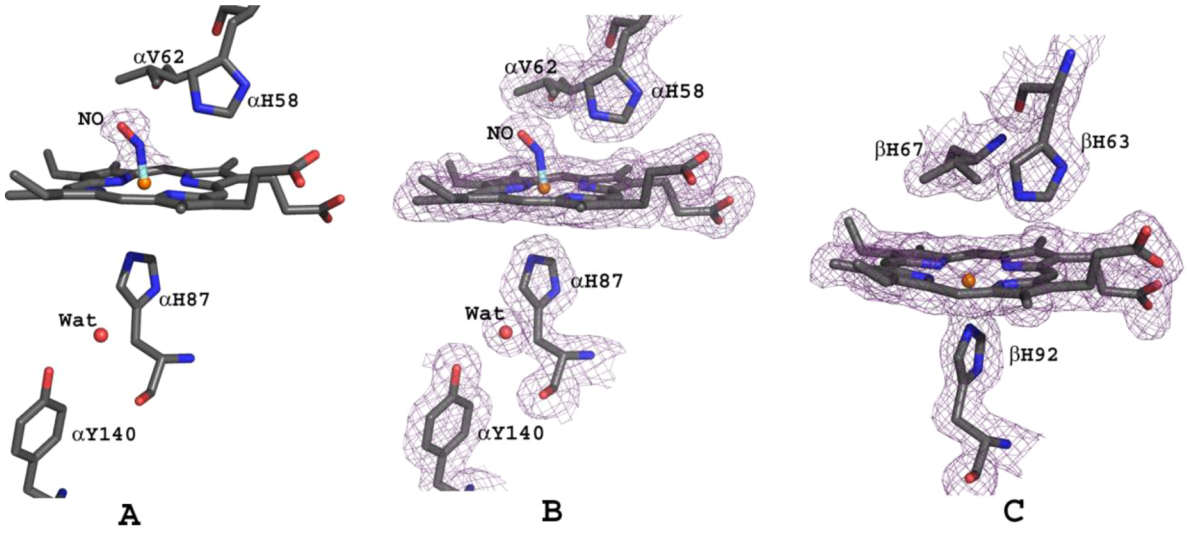


Figure 6. Heme environment of bis-ligated T-state HbNO. (A) Difference electron density ($F_o - F_c$) map of bound NO (before NO was built into the model) at the distal side of α1 heme, contoured at 3.5σ. Similar density is observed at the α2 heme. (B) Final $2F_o - F_c$ map of bound NO at the distal side of α1 heme, contoured at 1.2σ. (C) Final $2F_o - F_c$ map of β1 heme, contoured at 1.2σ. Similar density was observed at the β2 heme. The maps are superimposed with the final refined model.

The α Hemes but Not the β Hemes Are Coordinated with NO. Nitric oxide has been shown structurally to interact with the heme irons in both deoxyHb and oxyHb to form HbNO[48,49] and/or to interact with βCys93 of oxyHb to form HbSNO or HbSNOH.[53] In the structure reported here, NO likely produced via hydrolysis of DD-1 to RSR13 and/or RSR13-CH_2CH_2OH binds to the distal side of the two α heme irons, forming a bis-ligated T-state HbNO derivative structure (Figure 6A,B). It is also quite conceivable that, nitrite, as a result of the hydrolysis of DD-1 initially bound to the heme to form HbONO, was subsequently reduced to HbNO by sodium dithionite. A similar method has been used to obtain the nitrosyl derivative of Hb.[49,67] We found no structural evidence of a bound NO at βCys93 in our structure, which is consistent with modification of this residue only in the R-state where the cysteine is exposed.[50,68] Chan et al. recently published a T-state human HbNO structure with all four hemes coordinated with NO (tetra-ligated T-state HbNO);[48] however, unlike our bis-ligated T-state HbNO structure, where the crystal was obtained from solution, the tetra-ligated NO structure was obtained by exposing already formed T-state crystals to NO gas. A tetra-ligated R-state HbNO structure has also been reported by Richter-Addo and co-workers.[49] It is interesting that despite the high affinity of Hb for NO,[69,70] our structure showed ligation only at the two α hemes. The isolation of this bis-ligated HbNO structure, the first of its kind to be reported, likely occurred as a result of the bound RSR13 decreasing the affinity of the protein for the ligand, perhaps coupled with the fact that only a small quantity of NO might have been released during the hydrolysis of DD-1 as suggested by our NO release study.

In the published tetra-ligated T-state HbNO structure,[48] binding of NO has resulted in a complete breakage of the proximal Fe−N(His87) bond (4.06 Å) in the α hemes, but not in the β hemes. Interestingly, the corresponding α heme bond in our bis-ligated T-state HbNO is also broken but with a length of 3.3 Å between the iron and the proximal histidine as shown by the lack of overlapping density between the iron and the histidine (Figure 6B). In contrast, NO ligation of the α heme in the tetra-ligated R-state HbNO structure did not break the Fe−N(His87) bond (1.96 Å).[49] The broken or weakened Fe−N(His87) bonds in the tetra-ligated and bis-ligated T-state HbNO structures have resulted in the displacement of the αFe atom toward the distal side of the porphyrin (0.25 and 0.16 Å, respectively), while in the R-state structure, the iron is centered in the heme plane. These structural observations are consistent with Perutz's stereochemical theory that suggested that ligation of T-state Hb is opposed by tension at the α heme groups leading to rupture of the Fe−N(His87) bond.[71−74] As found in the tetra-ligated T-state HbNO structure, we also observed water-mediated hydrogen bond interactions between the proximal histidine and αTyr140[48] (Figure 6A,B), which has been proposed to stabilize the position of the cleaved Fe−N(His87) bond.[75] The Fe−N bond and Fe−N−O angle in the bis-ligated T-state HbNO were not restrained during refinement, resulting in an Fe−N distance of 1.7 Å, a N−O distance of 1.13 Å, and an Fe−N−O angle of 144°. This compares with values of 1.7 Å, 1.13 Å, and 138°, respectively, in the tetra-ligated T-state HbNO structure[48] and 1.8 Å, 1.18 Å, and 138°, respectively, in the tetra-ligated R-state HbNO structure.[49] The proximal αHis58 side chain makes 3.2 Å hydrogen bond interactions with both the oxygen and nitrogen of the NO ligand that compare with values of 3.35 and 3.4 Å in the tetra-ligated T-state HbNO structure and 2.7 and 2.8 Å in the tetra-ligated R-state HbNO structure, respectively.

Unlike the previous T-state and R-state HbNO structures that showed ligation of all hemes with NO, none of the β hemes in our structure is ligated with NO, and the structure remains five-coordinate with an Fe−N(His92) bond distance of 2.05 Å (Figure 6C), which compares with values of 2.25 and 2.0 Å for the tetra-ligated T-state and R-state HbNO structures, respectively. Expectedly, the βFe is displaced toward the distal side of the porphyrin plane by 0.15 Å in the bis-ligated T-state HbNO structure.

A hallmark of Hb allosterism is the position of the following heme environment structures: EF corner, F helix, and FG corner (Figure 7A,B). The T to R transition moves these structures toward the $\alpha1\beta2$ ($\alpha2\beta1$) dimer interface and/or the central water cavity, resulting in the relaxation of the strain at

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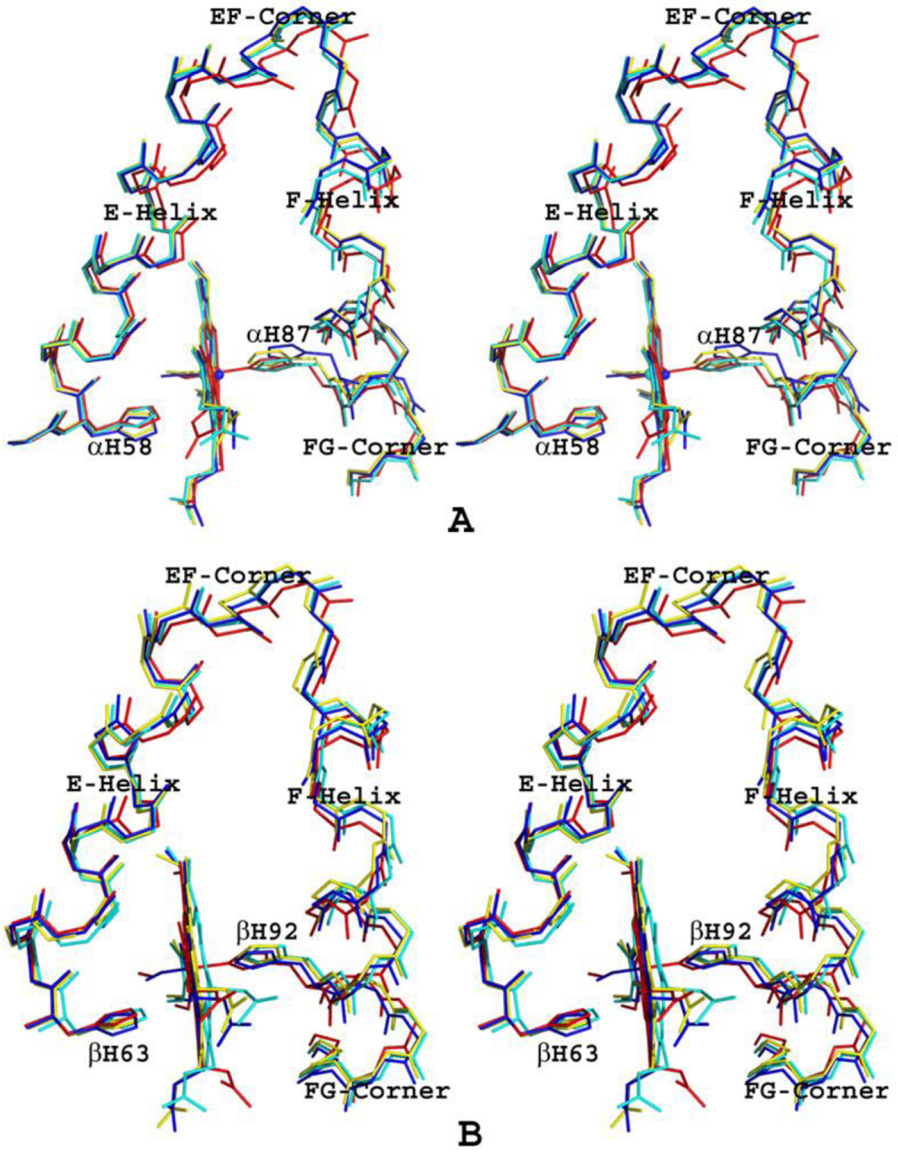


Figure 7. Stereodiagram of superposed heme environments of bis-ligated T-state HbNO (yellow), tetra-ligated T-state HbNO (blue), tetra-ligated R-state HbNO (red), and native T-state Hb (cyan): (A) α1 heme environment and (B) β1 heme environment.

the Fe−N(proximal histidine) bond with a concomitant increase in Hb oxygen affinity, and vice versa.[73,76] We note that in the α subunits of the tetra-ligated and bis-ligated HbNO structures, the EF corner (residues 71−74), F helix (residues 75−90), and FG corner (residues 91−97) are similarly positioned, but interestingly enough, even though the α hemes are ligated, these heme structures are further removed from their counterparts in the ligated R-state HbNO structure when compared to those of the unliganded native T-state structure (Figure 7B). This suggests a more tense α heme environment in the T-state HbNO structures than in the unliganded T-state structure, consistent with the observed ruptured Fe−N(His87) bonds in the liganded structures. A similar finding has been reported for the crystal structure of T-state cyanomethemoglobin, which is also characterized by a ruptured Fe−N(His87) bond.[71]

We also found significant differences in the positions of EF corner (residues 78 and 79), F helix (residues 80−95), and FG corner (residues 96−103) structures at the β heme. As expected, the EF corner, F helix, and FG corner of tetra-ligated R-state HbNO (with a bound β heme ligand) are the most relaxed. The EF corner, F helix, and FG corner of the tetra-ligated T-state HbNO (with a bound β heme ligand) are closer to their counterparts in the R-state HbNO structure, while those of bis-ligated T-state HbNO and native T-state crystal structures (without bound ligands at the β hemes) are close to each other but farthest from the R-state structure (Figure 7B).

Crystallographic lattice contacts in the tetra-ligated HbNO structure were proposed to be the stabilizing factor for preventing the crystalline T-state from transitioning to the R-state when the crystal was exposed to NO gas to form the nitrosylHb derivative.[48] In contrast, the bound RSR13 in our structure, as was previously proposed for inositol hexaphosphate (IHP)-bound nitrosylHb,[77] is providing the stabilization force for preventing the T to R transition with NO ligation. Because NO reacts extremely fast with heme[69,70] and dissociates very slowly,[78,79] it is not surprising that all four hemes of the previously reported HbNO crystal structures, including the T-state, are ligated with NO.[48,49] The significant

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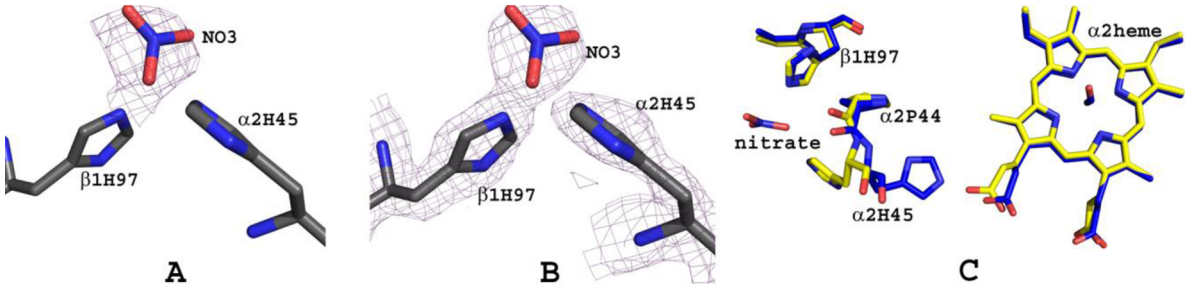


Figure 8. Nitrate binding at βHis97 of bis-ligated T-state HbNO. (A) Difference electron density ($F_o - F_c$) map of bound NO_3 (before NO_3 was built into the model), contoured at 3.5σ. (B) Final $2F_o - F_c$ map of bound NO_3 (before NO_3 was built into the model), contoured at 1.2σ. The maps were superimposed with the final refined model. (C) Comparison of the NO_3 binding site of bis-ligated T-state HbNO (yellow) and native T-state deoxyHb (blue).

presence of tetra-NO ligation Hb *in vivo* could be detrimental to the protein's primary function of binding and transporting oxygen; expectedly, under physiological conditions, only 0.1% of the total Hb is in the nitrosyl form.[80] As noted by Yonetani and co-workers, NO ligation of the heme *in vivo* appears to occur exclusively at the α hemes, allowing the nitrosylHb derivative to deliver oxygen to tissues as efficiently as native Hb.[81] Gladwin et al. also reported that Hb-mediated nitrite reduction proceeds at approximately the P_{50} where half the Hb is oxygen-bound.[82,83] Our bis-ligated T-state HbNO structure appears to capture for the first time the physiological nitrosylHb species, providing atomic-level insights into how partially NO-ligated Hb (at the α heme) may function to bind and transport oxygen. The exclusive ligation of the α heme in our structure is also consistent with sequential ligation of the heme, starting with the less sterically hindered α heme as proposed by Perutz.[73,74,76,84,85] Finally, the observation that binding of RSR13 to T-state Hb seems to prevent NO from coordinating to all four hemes has important medicinal implications as RSR13 could serve a useful purpose in decreasing the affinity of Hb for NO and thus increasing local NO bioavailability. In fact, RSR13 has been proposed as a potential agent to enhance the bioavailability of gases interacting with Hb, e.g., CO, NO, H_2S, etc., by reducing their level of binding with native hemoglobin and/or hemoglobin-based oxygen carriers (HBOCs).[86]

A Nitrate Molecule Binds at βHis97. Another novel finding of the crystallographic study is a bound nitrate at the two βHis97 side chains (Figure 8). The exact identity of the bound compound is uncertain; whether it is a nitrate ion or $HOCH_2CH_2ONO_2$, either could presumably be formed during the hydrolysis of DD-1. Each βHis97 binding site shows a strong electron density overlapping the imidazole of βHis97 that fits a four-atom nitrate but not large enough to fit $HOCH_2CH_2ONO_2$, necessitating modeling of NO_3 into the density (Figure 8A,B). One of the nitrate oxygens makes a strong hydrogen bond interaction with the side chain of βHis97 (~2.3 Å). Binding of the nitrate to βHis97 induces a conformational change in the vicinity of the binding site. In T-state structures, including the native, RSR13 T-state complex, tetra-ligated T-state HbNO, and classical R-state structure, the side chain of αHis45 makes strong intrasubunit hydrogen bond interaction with the α heme propionate (Figure 8C). The significance of this interaction, if any, has not been reported. Interestingly, in the bis-ligated T-state HbNO structure, this interaction is either broken with the imidazole of αHis45 rotated ~180° to make a hydrogen bond contact with the bound nitrate at the β1His97 site (Figure 8C) or weakened at the β2His97 site.

The question we pose is whether the bound nitrate is an artifact or perhaps reveals a possible role of βHis97 and αHis45 for nitrate regulation, providing an interesting alternative biological pathway of Hb preserving NO bioactivity. Similar adduct formation between βCys93 of oxyHb and NO[50,52] has been proposed, although not without controversy, as a way of conserving NO that is released from Hb to hypoxic tissue during the transition to the T-state.[68,87] Nitrated lipids[88−90] have also been proposed as NO storage and/or transport species, which eventually gets converted to NO. However, we could not find in the literature any mention or reported study of a nitrate binding to any specific Hb amino acid, although as noted by Thatcher et al., apart from the Fe heme site, Hb may also have a role in direct interaction with NO-containing compounds, including organic nitrites, nitrosothiols, and nitrates, providing the three electrons required for reduction of nitrate to NO.[60]

The Prodrugs Form Degradation Products in the Presence of Different Matrices: The Mechanism of Their Biotransformation. To gain insight into the biotransformation of the NO-releasing prodrugs, as well as their possible mechanism of pharmacologic activity, we used UPLC−MS analysis to follow their reaction products in the presence of different matrices (DMSO, buffer, L-cysteine, sodium dithionite, serum, bovine serum albumin, Hb, or whole blood) that mimic the various study conditions. Upon incubation with DMSO alone or buffer alone (50 mM Tris and 0.5 mM EDTA) or buffer with bovine serum albumin for 16 or 24 h, the compounds, including RSR13, showed no observable degradation, either at 37 °C (Figures S1 and S2) or at room temperature (not shown). When DD-1 was incubated in the presence of L-cysteine at 37 °C (mimicking the NO release study) for 24 h, we observed three main compounds, DD-1 (RSR13-$CH_2CH_2ONO_2$), RSR13-CH_2CH_2OH, and RSR13, that were present at levels of ~67, ~15, and ~18%, respectively (Figure S3). L-Cysteine is apparently hydrolyzing the carboxylate and nitrate ester groups of DD-1 to yield RSR13 and RSR13-CH_2CH_2OH, respectively, although most of the DD-1 remained intact. In contrast, DD-3 in the presence of L-cysteine showed two main compounds (Figure S3), corresponding to intact DD-3 (~60%) and RSR13 (~40%). Hydrolysis of the DD-3 carboxylate ester by L-cysteine appears to be more efficient than that of the DD-1 carboxylate ester.

Unlike L-cysteine, when DD-1 was incubated with human serum (containing nonspecific esterases) for 24 h at 37 °C, almost 100% of RSR13 was produced (Figure S4). DD-3 on the

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other hand yielded a mixture of unreacted DD-3 (~60%) and RSR13 (~40%) (Figure S4), the same ratio seen upon incubation with L-cysteine (Figure S3). It seems that esterases are more efficient in cleaving the carboxylate ester bond of the nitrooxyalkyl prodrug than that of the diazen-1-ium-1,2-diolate prodrug. Nevertheless, there appears to be no direct correlation between cleavage of the carboxylate ester (to form RSR13) by esterases and the amount of NO released, as DD-1 that was completely hydrolyzed to RSR13 released at most only 9% of NO while DD-3 that resulted in ~40% transformation to RSR13 released 41−50% NO. This observation suggests a different mechanism(s) of denitration for the two classes of compounds.

The crystal structure of deoxyHb cocrystallized with DD-1 showed DD-1, RSR13, NO, nitrate, and possibly RSR13-CH_2CH_2OH bound to the protein. We therefore used UPLC−MS analysis to follow the reaction products of DD-1 upon incubation with crystallization buffer [0.2 M sodium acetate trihydrate, 0.1 M sodium cacodylate trihydrate (pH 6.5), and 30% (w/v) PEG 8000] containing sodium dithionite at room temperature to determine whether we could observe similar degradation products. After incubation for 2 days (when crystals normally start to appear), we identified DD-1 (~73%), RSR13-CH_2CH_2OH (~18%), and RSR13 (~9%), and at 8 days, the mixture composition changed to DD-1 (~56%), RSR13-CH_2CH_2OH (~38%), and RSR13 (~6%) (Figure S5). Note that when sodium dithionite was removed from the reaction solution, DD-1 did not undergo hydrolysis, clearly indicating that sodium dithionite is responsible for both the carboxylate and nitrate ester hydrolyses to produce RSR13 and RSR13-CH_2CH_2OH, respectively, consistent with the structural findings. The large amount of DD-1 and small amount of RSR13 in the reaction mixture also confirm our proposition that the observed weak effector electron density is due to the low affinity of Hb for DD-1, as well as slow reaction between Hb and the prodrug.

We also followed the reaction products of DD-1 and DD-3 in whole blood at both room temperature and 37 °C. At room temperature, when DD-1 was incubated with whole blood for 16 h, we observed a mixture of DD-1 (~70%) and RSR13 (~30), and for DD-3, we also observed a mixture of DD-3 (~83%) and RSR13 (~17%) (Figure S6). At 37 °C, formation of RSR13 from both DD-1 and DD-3 significantly increased from ~30 to ~57% and from ~17 to ~83%, respectively (Figure S6). The result is consistent with the expected hydrolysis of the carboxylate ester of these compounds by esterases and perhaps other enzymes to form RSR13. Note that neither prodrug at room temperature or 37 °C showed any degradation product with DMSO alone, buffer, or BSA (Figures S1 and S2).

Finally, when DD-1 or DD-3 was incubated with clarified Hb (similar to the condition used for OEC analysis) for 16 h at room temperature, there was no apparent formation of the degradation product (Figure S7). This observation clearly suggests that the observed weak and delayed allosteric activity with clarified Hb is due to direct binding of the prodrugs to Hb and emphasizes the importance of hydrolysis of the prodrug into RSR13 to effect fast and significant allosteric activity.

It is clear that the nonbiological matrix, L-cysteine, and sodium dithionite act on both the carboxylate and nitrate esters of DD-1 to yield RSR13 and RSR13-CH_2CH_2OH; however, in the presence of the biological matrices, serum and whole blood, we observed only the final reaction product RSR13. DD-2 should behave like DD-1. DD-3 as expected from its chemical composition yielded only RSR13 in all matrices. On the basis of our results, and the previously proposed mechanism of biotransformation of NO-releasing prodrugs containing carboxylate and/or nitrate ester groups,[34,60] we expect the carboxylate ester bond of DD-1 and DD-2 to be hydrolyzed producing RSR13, and the hydroxyalkylnitrate group [$HO(CH_2)_nONO_2$, where n = 2 (DD-1) or 3 (DD-2)] and nitrite ion, the latter two compounds undergoing subsequent denitration reaction to release NO metabolites, including NO that may involve a series of enzymatic actions and/or direct $3e^-$ reduction.[60,91−98] The diazen-1-ium-1,2-diolate prodrug is also expected to undergo ester hydrolysis at the carboxylate group to release RSR13 and 2-(hydroxymethoxy)-1-(pyrrolidin-1-yl)diazene 1-oxide, the latter undergoing further dinitration to yield NO.[34]

CONCLUSIONS

RSR13 is an allosteric effector of Hb that decreases its affinity for O_2 and as a result allows the release of more O_2 to tissues.[5] A decreased level of NO is a cause of several pathological conditions, and increasing its local bioavailability has also found use in several ischemic-related pathologies because of its vasodilatory property.[28−36,99] We have synthesized a new group of NO-releasing prodrugs possessing either a nitrooxyalkyl or a 1-(pyrrolidin-1-yl)diazen-1-ium-1,2-diolate moiety attached to the carboxylate of RSR13. *In vitro* functional/biological studies showed that the compounds effectively released NO in serum and blood and also lowered the O_2 affinity of Hb, providing a potential dual mechanism of action that may be beneficial in the treatment of diseases with ischemic bases. Nonetheless, the study suggests the diazen-1-ium-1,2-diolate prodrugs of RSR13 to be potentially more superior than the nitrooxyalkyl analogues, which would have to be confirmed with animal model studies. The crystal structure of deoxygenated Hb cocrystallized with DD-1 showed the ester hydrolysis products, RSR13 and NO, bound to the central water cavity and α heme iron, respectively. Another interesting structural finding was a nitrate bound at βHis97, the relevance and significance of which are yet to be determined but provide the groundwork for future investigations. Overall, the study not only provides the feasibility of discovering potential therapeutic agents for the treatment of ischemia-related disease but also provides new insight into the biotransformation of NO-releasing prodrugs, offers an atomic-level understanding of how the compounds may function *in vivo* to effect their pharmacologic activities, and contributes to our understanding of Hb−NO biochemistry.

ASSOCIATED CONTENT

Supporting Information

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Additional data (PDF)

Accession Codes

The coordinates have been deposited as PDB entry 5E29.

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Notes

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■ ABBREVIATIONS

Hb, hemoglobin; OEC, oxygen equilibrium curve; RBC, red blood cell; deoxyHb, deoxygenated hemoglobin; COHb, carbonmonoxy hemoglobin; metHb, methemoglobin; NO, nitric oxide; HbNO, nitrosylHb; HbSNO, S-nitrosoHb; pO_2, partial pressure of oxygen; 2,3-DPG, 2,3-diphosphoglycerate; R-state, relaxed state; T-state, tense state; rmsd, root-mean-square deviation.

■ REFERENCES

(1) Back, T., Kohno, K., and Hossmann, K. A. (1994) Cortical negative DC deflections following middle cerebral artery occlusion and KCl-induced spreading depression: Effect on blood flow, tissue oxygenation, and electroencephalogram. *J. Cereb. Blood Flow Metab. 14*, 12−19.

(2) Safo, M. K., Ahmed, M. H., Ghatge, M. S., and Boyiri, T. (2011) Hemoglobin-ligand binding: Understanding hb function and allostery on atomic level. *Biochim. Biophys. Acta, Proteins Proteomics 1814*, 797−809.

(3) Poyart, C., Marden, M. C., and Kister, J. (1994) Bezafibrate derivatives as potent effectors of hemoglobin. *Methods Enzymol. 232*, 496−513.

(4) Liong, C., Ortiz, D., Ao-ieong, E., Navati, M. S., Friedman, J. M., and Cabrales, P. (2014) Localized increase of tissue oxygen tension by magnetic targeted drug delivery. *Nanotechnology 25*, 265102.

(5) Safo, M. K., Moure, C. M., Burnett, J. C., Joshi, G. S., and Abraham, D. J. (2001) High-resolution crystal structure of deoxy hemoglobin complexed with a potent allosteric effector. *Protein Sci. 10*, 951−957.

(6) Watson, J. C., Doppenberg, E. M., Bullock, M. R., Zauner, A., Rice, M. R., Abraham, D., Young, H. F., and Ulatowski, J. A. (1997) Effects of the allosteric modification of hemoglobin on brain oxygen and infarct size in a feline model of stroke. *Stroke 28*, 1624−1630.

(7) Suh, J. H., Stea, B., Nabid, A., Kresl, J. J., Fortin, A., Mercier, J. P., Senzer, N., Chang, E. L., Boyd, A. P., Cagnoni, P. J., and Shaw, E. (2006) Phase III study of efaproxiral as an adjunct to whole-brain radiation therapy for brain metastases. *J. Clin. Oncol. 24*, 106−114.

(8) Scott, C., Suh, J., Stea, B., Nabid, A., and Hackman, J. (2007) Improved survival, quality of life, and quality-adjusted survival in breast cancer patients treated with efaproxiral (efaproxyn) plus whole-brain radiation therapy for brain metastases. *Am. J. Clin. Oncol. 30*, 580−587.

(9) Wong, G. K., and Marsden, P. A. (1996) Nitric oxide synthases: Regulation in disease. *Nephrol., Dial., Transplant. 11*, 215−220.

(10) Lamas, S., Perez-Sala, D., and Moncada, S. (1998) Nitric oxide: From discovery to the clinic. *Trends Pharmacol. Sci. 19*, 436−438.

(11) Ignarro, L. J. (1999) Nitric oxide: A unique endogenous signaling molecule in vascular biology. *Biosci. Rep. 19*, 51−71.

(12) De Franceschi, L. (2009) Pathophisiology of sickle cell disease and new drugs for the treatment. *Mediterr. J. Hematol. Infect. Dis. 1*, e2009024.

(13) Bin, J. P., Doctor, A., Lindner, J., Hendersen, E. M., Elizabeth Le, D., Leong-Poi, H., Fisher, N. G., Christiansen, J., and Kaul, S. (2006) Effects of nitroglycerin on erythrocyte rheology and oxygen unloading: Novel role of S-nitrosohemoglobin in relieving myocardial ischemia. *Circulation 113*, 2502−2508.

(14) Muscara, M. N., and Wallace, J. L. (1999) Nitric oxide. V. therapeutic potential of nitric oxide donors and inhibitors. *Am. J. Physiol. 276*, G1313−6.

(15) Wallis, J. P. (2005) Nitric oxide and blood: A review. *Transfus. Med. 15*, 1−11.

(16) Liu, C., Wajih, N., Liu, X., Basu, S., Janes, J., Marvel, M., Keggi, C., Helms, C. C., Lee, A. N., Belanger, A. M., Diz, D. I., Laurienti, P. J., Caudell, D. L., Wang, J., Gladwin, M. T., and Kim-Shapiro, D. B. (2015) Mechanisms of human erythrocytic bioactivation of nitrite. *J. Biol. Chem. 290*, 1281−1294.

(17) McKnight, G. M., Smith, L. M., Drummond, R. S., Duncan, C. W., Golden, M., and Benjamin, N. (1997) Chemical synthesis of nitric oxide in the stomach from dietary nitrate in humans. *Gut 40*, 211−214.

(18) Benjamin, N., O'Driscoll, F., Dougall, H., Duncan, C., Smith, L., Golden, M., and McKenzie, H. (1994) Stomach NO synthesis. *Nature 368*, 502.

(19) Lundberg, J. O., and Weitzberg, E. (2005) NO generation from nitrite and its role in vascular control. *Arterioscler., Thromb., Vasc. Biol. 25*, 915−922.

(20) Lundberg, J. O., Weitzberg, E., Cole, J. A., and Benjamin, N. (2004) Nitrate, bacteria and human health. *Nat. Rev. Microbiol. 2*, 593−602.

(21) Weiner, D. L., and Brugnara, C. (2003) Hydroxyurea and sickle cell disease: A chance for every patient. *JAMA 289*, 1692−1694.

(22) Huang, J., Kim-Shapiro, D. B., and King, S. B. (2004) Catalase-mediated nitric oxide formation from hydroxyurea. *J. Med. Chem. 47*, 3495−3501.

(23) Rossaint, R., Falke, K. J., Lopez, F., Slama, K., Pison, U., and Zapol, W. M. (1993) Inhaled nitric oxide for the adult respiratory distress syndrome. *N. Engl. J. Med. 328*, 399−405.

(24) Sheridan, R. L., Zapol, W. M., Ritz, R. H., and Tompkins, R. G. (1999) Low-dose inhaled nitric oxide in acutely burned children with profound respiratory failure. *Surgery 126*, 856−862.

(25) Ichinose, F., Roberts, J. D., Jr, and Zapol, W. M. (2004) Inhaled nitric oxide: A selective pulmonary vasodilator: Current uses and therapeutic potential. *Circulation 109*, 3106−3111.

(26) Cabrales, P., Han, G., Nacharaju, P., Friedman, A. J., and Friedman, J. M. (2011) Reversal of hemoglobin-induced vaso-constriction with sustained release of nitric oxide. *Am. J. Physiol. Heart Circ. Physiol. 300*, H49−56.

(27) Nachuraju, P., Friedman, A. J., Friedman, J. M., and Cabrales, P. (2011) Exogenous nitric oxide prevents cardiovascular collapse during hemorrhagic shock. *Resuscitation 82*, 607−613.

(28) Ignarro, L. J., Lippton, H., Edwards, J. C., Baricos, W. H., Hyman, A. L., Kadowitz, P. J., and Gruetter, C. A. (1981) Mechanism of vascular smooth muscle relaxation by organic nitrates, nitrites, nitroprusside and nitric oxide: Evidence for the involvement of S-nitrosothiols as active intermediates. *J. Pharmacol. Exp. Ther. 218*, 739−749.

(29) Bates, J. N., Harrison, D. G., Myers, P. R., and Minor, R. L. (1991) EDRF: Nitrosylated compound or authentic nitric oxide. *Basic Res. Cardiol 86* (Suppl.2), 17−26.

(30) Stamler, J., Jaraki, O., Osborne, J., Simon, D., Keaney, J., Vita, J., and Loscalzo, J. (1992) S-nitroso-albumin is the predominant bioactive form of nitric-oxide in human plasma. *Circulation 86*, 757.

(31) Stamler, J. S., Jaraki, O., Osborne, J., Simon, D. I., Keaney, J., Vita, J., Singel, D., Valeri, C. R., and Loscalzo, J. (1992) Nitric oxide

DOI: 10.1021/acs.biochem.5b01074
Biochemistry 2015, 54, 7178−7192

circulates in mammalian plasma primarily as an S-nitroso adduct of serum albumin. *Proc. Natl. Acad. Sci. U. S. A. 89*, 7674−7677.
(32) Napoli, C., and Ignarro, L. J. (2003) Nitric oxide-releasing drugs. *Annu. Rev. Pharmacol. Toxicol. 43*, 97−123.
(33) Kaur, J., Bhardwaj, A., Huang, Z., Narang, D., Chen, T. Y., Plane, F., and Knaus, E. E. (2012) Synthesis and biological investigations of nitric oxide releasing nateglinide and meglitinide type II antidiabetic prodrugs: In-vivo antihyperglycemic activities and blood pressure lowering studies. *J. Med. Chem. 55*, 7883−7891.
(34) Velazquez, C., Praveen Rao, P. N., and Knaus, E. E. (2005) Novel nonsteroidal antiinflammatory drugs possessing a nitric oxide donor diazen-1-ium-1,2-diolate moiety: Design, synthesis, biological evaluation, and nitric oxide release studies. *J. Med. Chem. 48*, 4061−4067.
(35) Flaherty, J. T. (1992) Role of nitrates in acute myocardial infarction. *Am. J. Cardiol. 70*, B73−B81.
(36) Gruetter, D. Y., Gruetter, C. A., Barry, B. K., Baricos, W. H., Hyman, A. L., Kadowitz, P. J., and Ignarro, L. J. (1980) Activation of coronary arterial guanylate cyclase by nitric oxide, nitroprusside, and nitrosoguanidine–inhibition by calcium, lanthanum, and other cations, enhancement by thiols. *Biochem. Pharmacol. 29*, 2943−2950.
(37) Cosby, K., Partovi, K. S., Crawford, J. H., Patel, R. P., Reiter, C. D., Martyr, S., Yang, B. K., Waclawiw, M. A., Zalos, G., Xu, X., Huang, K. T., Shields, H., Kim-Shapiro, D. B., Schechter, A. N., Cannon, R. O., 3rd, and Gladwin, M. T. (2003) Nitrite reduction to nitric oxide by deoxyhemoglobin vasodilates the human circulation. *Nat. Med. 9*, 1498−1505.
(38) Gladwin, M. T., Schechter, A. N., Kim-Shapiro, D. B., Patel, R. P., Hogg, N., Shiva, S., Cannon, R. O., 3rd, Kelm, M., Wink, D. A., Espey, M. G., Oldfield, E. H., Pluta, R. M., Freeman, B. A., Lancaster, J. R., Jr, Feelisch, M., and Lundberg, J. O. (2005) The emerging biology of the nitrite anion. *Nat. Chem. Biol. 1*, 308−314.
(39) Curtis, E., Hsu, L. L., Noguchi, A. C., Geary, L., and Shiva, S. (2012) Oxygen regulates tissue nitrite metabolism. *Antioxid. Redox Signaling 17*, 951−961.
(40) Gladwin, M. T., and Kim-Shapiro, D. B. (2012) Vascular biology: Nitric oxide caught in traffic. *Nature 491*, 344−345.
(41) Gladwin, M. T., Shelhamer, J. H., Schechter, A. N., Pease-Fye, M. E., Waclawiw, M. A., Panza, J. A., Ognibene, F. P., and Cannon, R. O., 3rd. (2000) Role of circulating nitrite and S-nitrosohemoglobin in the regulation of regional blood flow in humans. *Proc. Natl. Acad. Sci. U. S. A. 97*, 11482−11487.
(42) Modin, A., Björne, H., Herulf, M., Alving, K., Weitzberg, E., and Lundberg, J. O. N. (2001) Nitrite-derived nitric oxide: A possible mediator of "acidic metabolic" vasodilation. *Acta Physiol. Scand. 171*, 9−16.
(43) Li, H., Samouilov, A., Liu, X., and Zweier, J. L. (2001) Characterization of the magnitude and kinetics of xanthine oxidase-catalyzed nitrite reduction: Evaluation of its role in nitric oxide generation in anoxic tissues. *J. Biol. Chem. 276*, 24482−24489.
(44) Dejam, A., Hunter, C. J., Pelletier, M. M., Hsu, L. L., Machado, R. F., Shiva, S., Power, G. G., Kelm, M., Gladwin, M. T., and Schechter, A. N. (2005) Erythrocytes are the major intravascular storage sites of nitrite in human blood. *Blood 106*, 734−739.
(45) Kelm, M., and Yoshida, K. (1996) in *Metabolic Fate of Nitric Oxide and Related N-Oxides* (Feelisch, M., and Stamler, J. S., Eds.) 1st ed., pp 47−58, Wiley, London.
(46) Wennmalm, A., Benthin, G., and Petersson, A. S. (1992) Dependence of the metabolism of nitric oxide (NO) in healthy human whole blood on the oxygenation of its red cell haemoglobin. *Br. J. Pharmacol. 106*, 507−508.
(47) Pietraforte, D., Mallozzi, C., Scorza, G., and Minetti, M. (1995) Role of thiols in the targeting of S-nitroso thiols to red blood cells. *Biochemistry 34*, 7177−7185.
(48) Chan, N. L., Kavanaugh, J. S., Rogers, P. H., and Arnone, A. (2004) Crystallographic analysis of the interaction of nitric oxide with quaternary-T human hemoglobin. *Biochemistry 43*, 118−132.
(49) Yi, J., Soares, A. S., and Richter-Addo, G. B. (2014) Crystallographic characterization of the nitric oxide derivative of R-state human hemoglobin. *Nitric Oxide 39*, 46−50.
(50) Jia, L., Bonaventura, C., Bonaventura, J., and Stamler, J. S. (1996) S-nitrosohaemoglobin: A dynamic activity of blood involved in vascular control. *Nature 380*, 221−226.
(51) Stamler, J. S., Jia, L., Eu, J. P., McMahon, T. J., Demchenko, I. T., Bonaventura, J., Gernert, K., and Piantadosi, C. A. (1997) Blood flow regulation by S-nitrosohemoglobin in the physiological oxygen gradient. *Science 276*, 2034−2037.
(52) Patel, R. P., Hogg, N., Spencer, N. Y., Kalyanaraman, B., Matalon, S., and Darley-Usmar, V. M. (1999) Biochemical characterization of human S-nitrosohemoglobin. effects on oxygen binding and transnitrosation. *J. Biol. Chem. 274*, 15487−15492.
(53) Chan, N. L., Rogers, P. H., and Arnone, A. (1998) Crystal structure of the S-nitroso form of liganded human hemoglobin. *Biochemistry 37*, 16459−16464.
(54) Safo, M. K., and Abraham, D. J. (2003) X-ray crystallography of hemoglobins. *Methods Mol. Med. 82*, 1−19.
(55) Abdulmalik, O., Safo, M. K., Chen, Q., Yang, J., Brugnara, C., Ohene-Frempong, K., Abraham, D. J., and Asakura, T. (2005) 5-hydroxymethyl-2-furfural modifies intracellular sickle haemoglobin and inhibits sickling of red blood cells. *Br. J. Haematol. 128*, 552−561.
(56) Safo, M. K., Abdulmalik, O., Danso-Danquah, R., Burnett, J. C., Nokuri, S., Joshi, G. S., Musayev, F. N., Asakura, T., and Abraham, D. J. (2004) Structural basis for the potent antisickling effect of a novel class of five-membered heterocyclic aldehydic compounds. *J. Med. Chem. 47*, 4665−4676.
(57) Brunger, A. T., Adams, P. D., Clore, G. M., DeLano, W. L., Gros, P., Grosse-Kunstleve, R. W., Jiang, J. S., Kuszewski, J., Nilges, M., Pannu, N. S., Read, R. J., Rice, L. M., Simonson, T., and Warren, G. L. (1998) Crystallography & NMR system: A new software suite for macromolecular structure determination. *Acta Crystallogr., Sect. D: Biol. Crystallogr. 54*, 905−921.
(58) Emsley, P., and Cowtan, K. (2004) Coot: Model-building tools for molecular graphics. *Acta Crystallogr., Sect. D: Biol. Crystallogr. 60*, 2126−2132.
(59) Tang, X., Xian, M., Trikha, M., Honn, K. V., and Wang, P. G. (2001) Synthesis of peptide-diazeniumdiolate conjugates: Towards enzyme activated antitumor agents. *Tetrahedron Lett. 42*, 2625−2629.
(60) Thatcher, G. R., Nicolescu, A. C., Bennett, B. M., and Toader, V. (2004) Nitrates and NO release: Contemporary aspects in biological and medicinal chemistry. *Free Radical Biol. Med. 37*, 1122−1143.
(61) Keefer, L. K. (2003) Progress toward clinical application of the nitric oxide-releasing diazeniumdiolates. *Annu. Rev. Pharmacol. Toxicol. 43*, 585−607.
(62) Csont, T., and Ferdinandy, P. (2005) Cardioprotective effects of glyceryl trinitrate: Beyond vascular nitrate tolerance. *Pharmacol. Ther. 105*, 57−68.
(63) Wang, E. Q., Balthasar, J. P., and Fung, H. L. (2004) Pharmacodynamics of in vivo nitroglycerin tolerance in normal conscious rats: Effects of dose and dosing protocol. *Pharm. Res. 21*, 114−120.
(64) Park, S. Y., Yokoyama, T., Shibayama, N., Shiro, Y., and Tame, J. R. (2006) 1.25 A resolution crystal structures of human haemoglobin in the oxy, deoxy and carbonmonoxy forms. *J. Mol. Biol. 360*, 690−701.
(65) Phelps Grella, M., Danso-Danquah, R., Safo, M. K., Joshi, G. S., Kister, J., Marden, M., Hoffman, S. J., and Abraham, D. J. (2000) Synthesis and structure-activity relationships of chiral allosteric modifiers of hemoglobin. *J. Med. Chem. 43*, 4726−4737.
(66) Youssef, A. M., Safo, M. K., Danso-Danquah, R., Joshi, G. S., Kister, J., Marden, M. C., and Abraham, D. J. (2002) Synthesis and X-ray studies of chiral allosteric modifiers of hemoglobin. *J. Med. Chem. 45*, 1184−1195.
(67) Cheng, L., and Richter-Addo, G. B. (2000) Binding and Activation of Nitric Oxide by Metalloporphyrins and Heme. In *The Porphyrin Handbook* (Guilard, R., Smith, K., and Kadish, K. M., Eds.) Vol. *4*, pp 219−291, Academic Press, New York.

DOI: 10.1021/acs.biochem.5b01074
Biochemistry 2015, 54, 7178−7192

(68) McMahon, T. J., Exton Stone, A., Bonaventura, J., Singel, D. J., and Solomon Stamler, J. (2000) Functional coupling of oxygen binding and vasoactivity in S-nitrosohemoglobin. *J. Biol. Chem. 275*, 16738−16745.
(69) Cassoly, R., and Gibson, Q. (1975) Conformation, co-operativity and ligand binding in human hemoglobin. *J. Mol. Biol. 91*, 301−313.
(70) Kim-Shapiro, D. B. (2004) Hemoglobin-nitric oxide cooperativity: Is NO the third respiratory ligand? *Free Radical Biol. Med. 36*, 402−412.
(71) Paoli, M., Dodson, G., Liddington, R. C., and Wilkinson, A. J. (1997) Tension in haemoglobin revealed by fe-his(F8) bond rupture in the fully liganded T-state. *J. Mol. Biol. 271*, 161−167.
(72) Perutz, M. F. (1972) Stereochemical mechanism of cooperative effects in haemoglobin. *Biochimie 54*, 587−588.
(73) Perutz, M. F. (1972) Nature of haem-haem interaction. *Nature 237*, 495−499.
(74) Perutz, M. F., Wilkinson, A. J., Paoli, M., and Dodson, G. G. (1998) The stereochemical mechanism of the cooperative effects in hemoglobin revisited. *Annu. Rev. Biophys. Biomol. Struct. 27*, 1−34.
(75) Petruk, A. A., Vergara, A., Estrin, D., and Merlino, A. (2013) Molecular basis of the NO trans influence in quaternary T-state human hemoglobin: A computational study. *FEBS Lett. 587*, 2393−2398.
(76) Perutz, M. F., and TenEyck, L. F. (1972) Stereochemistry of cooperative effects in hemoglobin. *Cold Spring Harbor Symp. Quant. Biol. 36*, 295−310.
(77) Maxwell, J. C., and Caughey, W. S. (1976) An infrared study of NO bonding to heme B and hemoglobin A. evidence for inositol hexaphosphate induced cleavage of proximal histidine to iron bonds. *Biochemistry 15*, 388−396.
(78) Moore, E. G., and Gibson, Q. H. (1976) Cooperativity in the dissociation of nitric oxide from hemoglobin. *J. Biol. Chem. 251*, 2788−2794.
(79) Azizi, F., Kielbasa, J. E., Adeyiga, A. M., Maree, R. D., Frazier, M., Yakubu, M., Shields, H., King, S. B., and Kim-Shapiro, D. B. (2005) Rates of nitric oxide dissociation from hemoglobin. *Free Radical Biol. Med. 39*, 145−151.
(80) Angelo, M., Hausladen, A., Singel, D. J., and Stamler, J. S. (2008) Interactions of NO with hemoglobin: From microbes to man. *Methods Enzymol. 436*, 131−168.
(81) Yonetani, T., Tsuneshige, A., Zhou, Y., and Chen, X. (1998) Electron paramagnetic resonance and oxygen binding studies of alpha-nitrosyl hemoglobin. A novel oxygen carrier having no-assisted allosteric functions. *J. Biol. Chem. 273*, 20323−20333.
(82) Duranski, M. R., Greer, J. J., Dejam, A., Jaganmohan, S., Hogg, N., Langston, W., Patel, R. P., Yet, S. F., Wang, X., Kevil, C. G., Gladwin, M. T., and Lefer, D. J. (2005) Cytoprotective effects of nitrite during in vivo ischemia-reperfusion of the heart and liver. *J. Clin. Invest. 115*, 1232−1240.
(83) Huang, Z., Shiva, S., Kim-Shapiro, D. B., Patel, R. P., Ringwood, L. A., Irby, C. E., Huang, K. T., Ho, C., Hogg, N., Schechter, A. N., and Gladwin, M. T. (2005) Enzymatic function of hemoglobin as a nitrite reductase that produces NO under allosteric control. *J. Clin. Invest. 115*, 2099−2107.
(84) Perutz, M. F., Pulsinelli, P. D., and Ranney, H. M. (1972) Structure and subunit interaction of haemoglobin M milwaukee. *Nat. New Biol. 237*, 259−263.
(85) Perutz, M. F. (1989) Mechanisms of cooperativity and allosteric regulation in proteins. *Q. Rev. Biophys. 22*, 139−237.
(86) Safo, M., and Ward, K. R. (2012) Use of hemoglobin effectors to increase the bioavailability of therapeutic gases. WO2012075244 A2.
(87) Hobbs, A. J., Gladwin, M. T., Patel, R. P., Williams, D. L., and Butler, A. R. (2002) Haemoglobin: NO transporter, NO inactivator or NOne of the above? *Trends Pharmacol. Sci. 23*, 406−411.
(88) Schopfer, F. J., Baker, P. R., Giles, G., Chumley, P., Batthyany, C., Crawford, J., Patel, R. P., Hogg, N., Branchaud, B. P., Lancaster, J. R., Jr, and Freeman, B. A. (2005) Fatty acid transduction of nitric oxide signaling. nitrolinoleic acid is a hydrophobically stabilized nitric oxide donor. *J. Biol. Chem. 280*, 19289−19297.
(89) Baker, P. R., Schopfer, F. J., Sweeney, S., and Freeman, B. A. (2004) Red cell membrane and plasma linoleic acid nitration products: Synthesis, clinical identification, and quantitation. *Proc. Natl. Acad. Sci. U. S. A. 101*, 11577−11582.
(90) Lim, D. G., Sweeney, S., Bloodsworth, A., White, C. R., Chumley, P. H., Krishna, N. R., Schopfer, F., O'Donnell, V. B., Eiserich, J. P., and Freeman, B. A. (2002) Nitrolinoleate, a nitric oxide-derived mediator of cell function: Synthesis, characterization, and vasomotor activity. *Proc. Natl. Acad. Sci. U. S. A. 99*, 15941−15946.
(91) Gruetter, C. A., Barry, B. K., McNamara, D. B., Kadowitz, P. J., and Ignarro, L. J. (1980) Coronary arterial relaxation and guanylate cyclase activation by cigarette smoke, N′-nitrosonornicotine and nitric oxide. *J. Pharmacol. Exp. Ther. 214*, 9−15.
(92) McGuire, J. J., Anderson, D. J., McDonald, B. J., Narayanasami, R., and Bennett, B. M. (1998) Inhibition of NADPH-cytochrome P450 reductase and glyceryl trinitrate biotransformation by diphenyleneiodonium sulfate. *Biochem. Pharmacol. 56*, 881−893.
(93) Sampson, J. B., Ye, Y., Rosen, H., and Beckman, J. S. (1998) Myeloperoxidase and horseradish peroxidase catalyze tyrosine nitration in proteins from nitrite and hydrogen peroxide. *Arch. Biochem. Biophys. 356*, 207−213.
(94) van der Vliet, A., Eiserich, J. P., Halliwell, B., and Cross, C. E. (1997) Formation of reactive nitrogen species during peroxidase-catalyzed oxidation of nitrite. A potential additional mechanism of nitric oxide-dependent toxicity. *J. Biol. Chem. 272*, 7617−7625.
(95) Gladwin, M. T., Crawford, J. H., and Patel, R. P. (2004) The biochemistry of nitric oxide, nitrite, and hemoglobin: Role in blood flow regulation. *Free Radical Biol. Med. 36*, 707−717.
(96) Ignarro, L. J., Barry, B. K., Gruetter, D. Y., Edwards, J. C., Ohlstein, E. H., Gruetter, C. A., and Baricos, W. H. (1980) Guanylate cyclase activation of nitroprusside and nitrosoguanidine is related to formation of S-nitrosothiol intermediates. *Biochem. Biophys. Res. Commun. 94*, 93−100.
(97) Ignarro, L. J., and Gruetter, C. A. (1980) Requirement of thiols for activation of coronary arterial guanylate cyclase by glyceryl trinitrate and sodium nitrite: Possible involvement of S-nitrosothiols. *Biochim. Biophys. Acta, Gen. Subj. 631*, 221−231.
(98) Ignarro, L. J., Edwards, J. C., Gruetter, D. Y., Barry, B. K., and Gruetter, C. A. (1980) Possible involvement of S-nitrosothiols in the activation of guanylate cyclase by nitroso compounds. *FEBS Lett. 110*, 275−278.
(99) Fox-Robichaud, A., Payne, D., Hasan, S. U., Ostrovsky, L., Fairhead, T., Reinhardt, P., and Kubes, P. (1998) Inhaled NO as a viable antiadhesive therapy for ischemia/reperfusion injury of distal microvascular beds. *J. Clin. Invest. 101*, 2497−2505.

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